
Theory of Probability

Probability & Random Processes — MAT303

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Consolidated Lecture Notes

Definitions → Results → Problems → Solutions

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Contents

1	Introduction and Basic Terminology	3
2	Different Kinds of Events and the Addition Theorem	4
3	Conditional Probability and the Multiplication Theorem	5
4	Independent Events	6
5	Problems on Independence and the Multiplication Theorem	8
6	Bayes' Theorem	12
	Quick Reference	16

1 Introduction and Basic Terminology

A Short History

Historical Background

- A gambler's dispute in **1654** led to the creation of a mathematical theory of probability by two famous French mathematicians, **Blaise Pascal** and **Pierre de Fermat**.
- The first book on probability was written by the Dutch scientist **Christian Huygens** in **1657**.
- In **1933** the Russian mathematician **A. Kolmogorov** outlined an axiomatic approach that forms the basis for the modern theory.
- Applications occur in widely different fields such as genetics, psychology, economics and engineering.

Where do we use the word “Probability”?

Definition 1.1 (Stochastic Experiment)

Because of the stochastic nature of genetics and evolution we have to rely on the theory of probability. Whenever we are *unsure* about the outcome of an event, we can talk about the *probabilities* of certain outcomes.

Examples.

- Number of daily confirmed cases of COVID-19 patients.
- Detection of an aircraft in a radar range.

Terminologies

Definition 1.2 (Sample Space S)

The set of **all possible outcomes** of an experiment.

Examples.

- Number of daily confirmed cases of COVID-19 patients: $S = \{0, 1, 2, \dots, 5000\}$.
- Detection of an aircraft in a radar range: $S = \{v(t), v(t) + X_t\}$.

Definition 1.3 (Events $A, B, C, \dots$)

Elements of the sample space satisfying a common criterion.

Example. In a sample space of all patients in a hospital,

- A — patients reported COVID-19 positive,
- B — patients having heart diseases,
- C — patients having kidney diseases, and so on.

Definition 1.4 (Probability of an Event)

It is the *chance of happening* of the event, defined by

$$P(A) = \frac{\text{Number of cases favourable to } A}{\text{Total number of likely cases}} = \frac{n(A)}{n(S)}.$$

Axiomatic Definition**Definition 1.5 (Kolmogorov's Axioms)**

- | | |
|-------|-------------------------|
| (i) | $0 \leq P(A_i) \leq 1$ |
| (ii) | $P(S) = 1$ |
| (iii) | $P(\phi) = 0$ |
| (iv) | $P(\bar{A}) = 1 - P(A)$ |

Set-Theoretical Notations

We can use set-theoretical notations to combine more than one event:

Notation	Meaning
$A \cup B$	happening of either A or B
$A \cap B$	happening of both A and B
$A \cap \bar{B}$	happening of A and non-happening of B
$\bar{A} \cap \bar{B}$	happening of neither A nor B
$A \cup \bar{B}$	$= \bar{A} \cap \bar{B}$ (De Morgan), and so on

2 Different Kinds of Events and the Addition Theorem**Definition 2.1 (Mutually Exclusive Events – MEE)**

Two events are **mutually exclusive** if they *cannot happen simultaneously*, i.e. if they do not have common elements.

Example. A person is either COVID-19 positive or negative — not both.

Result 2.1 (Consequences for MEE)

If A and B are mutually exclusive, then

$$A \cap B = \phi \implies P(A \cap B) = 0, \quad \text{and} \quad P(A \cup B) = P(A) + P(B).$$

Result 2.2 (Addition Theorem – any two events)

For *any* two events A and B ,

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

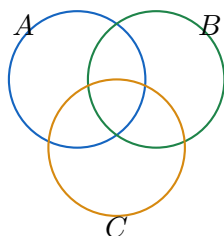
Result 2.3 (Addition Theorem – any three events)

For any three events A , B and C ,

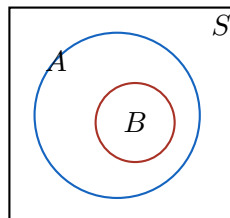
$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(C \cap A) + P(A \cap B \cap C).$$

Result 2.4 (Monotonicity)

If $B \subseteq A$ then $P(B) \leq P(A)$.



(a) Any three events

(b) $B \subseteq A$ **Problem 1 (Addition Theorem)**

In a metropolitan city it is noticed that the probability of people affected by COVID-19 is 0.6, with heart problem is 0.3, and the probability that people are affected by both COVID-19 as well as heart problem is 0.3. Identify the probability that

- (i) people are affected by COVID-19 only and not by heart problem;
- (ii) people are affected by neither of the diseases.

Solution 1

Let A = “affected by COVID-19” and B = “affected by heart problem”. Given

$$P(A) = 0.6, \quad P(B) = 0.3, \quad P(A \cap B) = 0.3.$$

- (i) COVID-19 only, not heart problem:

$$P(A \cap \overline{B}) = P(A) - P(A \cap B) = 0.6 - 0.3 = 0.3$$

- (ii) Neither of the diseases. First,

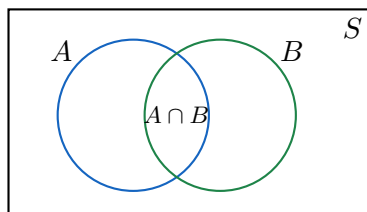
$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.6 + 0.3 - 0.3 = 0.6.$$

Hence, by De Morgan’s law,

$$P(\overline{A \cap B}) = P(\overline{A \cup B}) = 1 - P(A \cup B) = 1 - 0.6 = 0.4$$

3 Conditional Probability and the Multiplication Theorem

Motivation



Suppose the event A has *already happened*. Then the sample space effectively shrinks from S to A , and the favourable outcomes for B are only those inside $A \cap B$. Hence

$$P(B / A) = \frac{n(A \cap B)}{n(A)} = \frac{n(A \cap B)/n(S)}{n(A)/n(S)} = \frac{P(A \cap B)}{P(A)}.$$

Definition 3.1 (Conditional Probability)

$$P(B / A) = \frac{P(A \cap B)}{P(A)}, \quad P(A) \neq 0$$

and similarly

$$P(A / B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) \neq 0.$$

Result 3.1 (Multiplication Theorem in Probability)

Combining the two definitions above,

$$P(A \cap B) = P(A) \cdot P(B / A) = P(B) \cdot P(A / B)$$

and, for three events,

$$P(A \cap B \cap C) = P(A) \cdot P(B / A) \cdot P(C / A \cap B).$$

Result 3.2 (Special Cases)

(1) If $B \subseteq A$ then $A \cap B = B$, so

$$P(A / B) = \frac{P(A \cap B)}{P(B)} = \frac{P(B)}{P(B)} = 1.$$

(2) If A and B are two mutually exclusive events then $P(A / B) = P(B / A) = 0$.

4 Independent Events

Definition 4.1 (Independent Events)

Starting from the multiplication theorem $P(A \cap B) = P(A) \cdot P(B / A)$: if the occurrence of A gives *no information* about B , then $P(B / A) = P(B)$, and we obtain

$$A \text{ and } B \text{ are independent} \iff P(A \cap B) = P(A) \cdot P(B).$$

Result 4.1 (Complements are also independent)

If A and B are independent events, then $\bar{A}$ and $\bar{B}$ are also independent events.

Proof of Result 4.1

Given A and B are independent events, so that

$$P(A \cap B) = P(A) \cdot P(B). \quad (1)$$

To prove: $\bar{A}$ and $\bar{B}$ are also independent events. Now

$$\begin{aligned} P(\bar{A} \cap \bar{B}) &= P(\overline{A \cup B}) & (\because \text{De Morgan's law}) \\ &= 1 - P(A \cup B) \\ &= 1 - (P(A) + P(B) - P(A \cap B)) \\ &= 1 - P(A) - P(B) + P(A) \cdot P(B) & (\because \text{by (1)}) \\ &= 1 - P(A) - P(B) (1 - P(A)) \\ &= (1 - P(A))(1 - P(B)) \\ &= P(\bar{A}) \cdot P(\bar{B}). \end{aligned}$$

Hence $\bar{A}$ and $\bar{B}$ are also independent events. ■

Result 4.2 (Mixed complements)

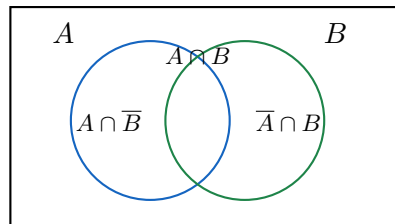
If A and B are independent events, then $\bar{A}$ and B are independent; similarly A and $\bar{B}$ are independent.

Proof of Result 4.2

Given A and B are independent, so $P(A \cap B) = P(A) \cdot P(B)$. From the Venn diagram, B splits into the disjoint pieces $A \cap B$ and $\bar{A} \cap B$. Hence

$$\begin{aligned} P(\bar{A} \cap B) &= P(B) - P(A \cap B) \\ &= P(B) - P(A) \cdot P(B) \\ &= P(B) (1 - P(A)) \\ &= P(B) \cdot P(\bar{A}). \end{aligned}$$

Therefore $\bar{A}$ and B are also independent events. Similarly, $P(A \cap \bar{B}) = P(A) \cdot P(\bar{B})$, so A and $\bar{B}$ are also independent events. ■



Result 4.3 (Extension to three independent events)

If A , B and C are independent events, then

$$P(\overline{A} \cap \overline{B} \cap \overline{C}) = P(\overline{A}) \cdot P(\overline{B}) \cdot P(\overline{C}).$$

Summary of the Basic Formulae

Situation	Formula
Mutually exclusive events (MEE)	$P(A \cup B) = P(A) + P(B)$ $P(A \cap B) = 0$
Any two events	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$ (addition)
Conditional events	$P(A/B) = \frac{P(A \cap B)}{P(B)}, \quad P(B/A) = \frac{P(A \cap B)}{P(A)}$ $P(A \cap B) = P(A) \cdot P(B/A)$ (multiplication)
Independent events	$P(A \cap B) = P(A) \cdot P(B)$ $P(\overline{A} \cap \overline{B}) = P(\overline{A}) \cdot P(\overline{B})$ $P(\overline{A} \cap B) = P(\overline{A}) \cdot P(B)$

5 Problems on Independence and the Multiplication Theorem**Problem 2 (Three students solving a problem)**

A problem in Statistics is given to three students A , B and C whose chances of solving the problem are $1/2$, $3/4$ and $1/4$ respectively. What is the probability that

- (i) no one will solve the problem;
- (ii) only one will solve the problem;
- (iii) the problem will be solved?

Solution 2

Given

$$P(A) = \frac{1}{2}, \quad P(B) = \frac{3}{4}, \quad P(C) = \frac{1}{4},$$

and solving the problem by one student is *independent* of the others. Hence

$$P(\overline{A}) = 1 - P(A) = \frac{1}{2}, \quad P(\overline{B}) = 1 - \frac{3}{4} = \frac{1}{4}, \quad P(\overline{C}) = 1 - \frac{1}{4} = \frac{3}{4}.$$

(i) No one will solve the problem.

$$\begin{aligned}
 P(\text{no one solves}) &= P(\overline{A} \cap \overline{B} \cap \overline{C}) \\
 &= P(\overline{A}) \cdot P(\overline{B}) \cdot P(\overline{C}) \quad (\because \text{they are independent}) \\
 &= \frac{1}{2} \cdot \frac{1}{4} \cdot \frac{3}{4} = \frac{3}{32}
 \end{aligned}$$

(ii) Only one will solve the problem.

$$\begin{aligned}
 P(\text{only one solves}) &= P((A \cap \bar{B} \cap \bar{C}) \cup (\bar{A} \cap B \cap \bar{C}) \cup (\bar{A} \cap \bar{B} \cap C)) \\
 &= P(A \cap \bar{B} \cap \bar{C}) + P(\bar{A} \cap B \cap \bar{C}) + P(\bar{A} \cap \bar{B} \cap C) \quad (\because \text{MEE}) \\
 &= P(A)P(\bar{B})P(\bar{C}) + P(\bar{A})P(B)P(\bar{C}) + P(\bar{A})P(\bar{B})P(C) \\
 &= \frac{1}{2} \cdot \frac{1}{4} \cdot \frac{3}{4} + \frac{1}{2} \cdot \frac{3}{4} \cdot \frac{3}{4} + \frac{1}{2} \cdot \frac{1}{4} \cdot \frac{1}{4} \\
 &= \frac{3}{32} + \frac{9}{32} + \frac{1}{32} = \boxed{\frac{13}{32}}
 \end{aligned}$$

(iii) The problem will be solved.

$$\begin{aligned}
 P(\text{problem is solved}) &= P(A \cup B \cup C) = 1 - P(\overline{A \cup B \cup C}) = 1 - P(\bar{A} \cap \bar{B} \cap \bar{C}) \\
 &= 1 - \frac{3}{32} = \boxed{\frac{29}{32}}
 \end{aligned}$$

Problem 3 (Drawing without replacement)

A lot contains 10 items out of which three are defective. Three items are chosen from the lot one after another at random *without replacement*. Find the probability that all the three are defective.

Solution 3

10 items $\longrightarrow$ 3 defective

Three items are taken one after another *without replacement*; let the corresponding events be A , B and C (first, second and third item drawn is defective). Since the items are not replaced, the events are *dependent*, so we use the multiplication theorem:

$$\begin{aligned}
 P(\text{all three drawn are defective}) &= P(A \cap B \cap C) \\
 &= P(A) \times P(B / A) \times P(C / A \cap B) \\
 &= \frac{3}{10} \times \frac{2}{9} \times \frac{1}{8} = \boxed{\frac{1}{120}}
 \end{aligned}$$

Problem 4 (Last defective on the ninth test)

A consignment of 15 items contains 4 defectives. The items are selected at random one by one and tested; the tested items are *not* put back. What is the probability that the 9th one tested is the last defective?

Solution 4

Let

- A — the event of getting 3 defective items out of the first 8 tested;
- B — the event of getting a defective item on the 9th test.

The 9th item tested is the *last* defective precisely when exactly 3 defectives appear in the first 8 tests and the 9th is the remaining defective.

Choosing 3 defectives from 4 and 5 non-defectives from 11:

$$P(A) = \frac{{}^4C_3 \times {}^{11}C_5}{{}^{15}C_8} = \frac{\frac{4!}{1!3!} \times \frac{11!}{6!5!}}{\frac{15!}{7!8!}} = \frac{4 \times 462}{6435} = \frac{56}{195} \quad \left(\because {}^nC_r = \frac{n!}{(n-r)!r!} \right)$$

Since the tested items are not replaced, after 8 tests there remain 7 items containing exactly 1 defective. Therefore

$$P(B / A) = \frac{1}{7}.$$

Hence

$$\begin{aligned} P(\text{the 9th one examined is the last defective}) &= P(A \cap B) \\ &= P(A) \times P(B / A) \\ &= \frac{56}{195} \times \frac{1}{7} = \boxed{\frac{8}{195}} \end{aligned}$$

Problem 5 (At least one chip of each colour)

A box contains 6 red, 4 white and 5 black coloured chips. A person draws 4 chips from the box at random. Find the probability that among the chips drawn there is at least one chip of each colour.

Solution 5

The box contains

$$6R + 4W + 5B = 15 \text{ chips,}$$

and 4 chips are drawn at random. To have *at least one of each colour* among the 4 chips, the only possible splits are

$$(2R, 1W, 1B), \quad (1R, 2W, 1B), \quad (1R, 1W, 2B).$$

These three cases are mutually exclusive, so

$$\begin{aligned} P(\text{at least one chip of each colour}) &= P((2R, 1W, 1B) \cup (1R, 2W, 1B) \cup (1R, 1W, 2B)) \\ &= P(2R, 1W, 1B) + P(1R, 2W, 1B) + P(1R, 1W, 2B) \\ &= \frac{{}^6C_2 \times {}^4C_1 \times {}^5C_1}{{}^{15}C_4} + \frac{{}^6C_1 \times {}^4C_2 \times {}^5C_1}{{}^{15}C_4} + \frac{{}^6C_1 \times {}^4C_1 \times {}^5C_2}{{}^{15}C_4} \\ &= \frac{1}{{}^{15}C_4} [(15)(4)(5) + (6)(6)(5) + (6)(4)(10)] \\ &= \frac{300 + 180 + 240}{1365} = \frac{720}{1365} = \boxed{\frac{48}{91}} \end{aligned}$$

Problem 6 (Alternate throws with a pair of dice)

Let A and B throw alternately with a pair of dice. A wins if he throws 6 before B throws 7, and B wins if he throws 7 before A throws 6. If A begins the game, show that his chance of winning is $30/61$.

Solution 6

Let A be the event of getting a total of 6 when a pair of dice is thrown. Then

$$A = \{(1, 5), (2, 4), (3, 3), (4, 2), (5, 1)\} \implies P(\text{getting } 6) = P(A) = \frac{5}{36},$$

$$P(\text{not getting } 6) = P(\overline{A}) = 1 - \frac{5}{36} = \frac{31}{36}.$$

Similarly, let B be the event of getting a total of 7. Then

$$B = \{(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)\} \implies P(\text{getting } 7) = P(B) = \frac{6}{36} = \frac{1}{6},$$

$$P(\overline{B}) = 1 - \frac{1}{6} = \frac{5}{6}.$$

A wins if he throws 6 before B throws 7, and A starts the game. Writing the suffix for the *chance number* (i.e. A_k = “ A throws 6 on his k th turn”),

$P(\text{winning chance of } A)$

$$= P((\text{1st chance}) \cup (\text{2nd chance}) \cup (\text{3rd chance}) \cup \dots)$$

$$= P(A_1 \cup (\overline{A}_1 \cap \overline{B}_1 \cap A_2) \cup (\overline{A}_1 \cap \overline{B}_1 \cap \overline{A}_2 \cap \overline{B}_2 \cap A_3) \cup \dots)$$

$$= P(A_1) + P(\overline{A}_1 \cap \overline{B}_1 \cap A_2) + P(\overline{A}_1 \cap \overline{B}_1 \cap \overline{A}_2 \cap \overline{B}_2 \cap A_3) + \dots \quad (\because \text{they are MEE})$$

$$= P(A_1) + P(\overline{A}_1) P(\overline{B}_1) P(A_2) + P(\overline{A}_1) P(\overline{B}_1) P(\overline{A}_2) P(\overline{B}_2) P(A_3) + \dots$$

$$= \frac{5}{36} + \left(\frac{31}{36} \times \frac{5}{6} \times \frac{5}{36}\right) + \left(\frac{31}{36} \times \frac{5}{6} \times \frac{31}{36} \times \frac{5}{6} \times \frac{5}{36}\right) + \dots$$

$$= \frac{5}{36} \left[1 + \left(\frac{31}{36} \times \frac{5}{6}\right) + \left(\frac{31}{36} \times \frac{5}{6}\right)^2 + \dots\right]$$

$$= \frac{5}{36} (1 + x + x^2 + \dots) = \frac{5}{36} (1 - x)^{-1}, \quad \text{where } x = \frac{31}{36} \times \frac{5}{6} = \frac{155}{216}.$$

Therefore

$$P(\text{winning chance of } A) = \frac{5}{36} \left(1 - \frac{155}{216}\right)^{-1} = \frac{5}{36} \left(\frac{61}{216}\right)^{-1} = \frac{5}{36} \times \frac{216}{61} = \frac{30}{61}$$

Winning chance of B

Since only two players are playing the game,

$$P(\text{winning chance of } B) = 1 - P(\text{winning chance of } A) = 1 - \frac{30}{61} = \frac{31}{61}.$$

Problem 7 (Three players tossing a coin)

A , B and C in order toss a coin, and whoever throws the first head wins the game. Find their respective chances of winning.

Solution 7

For a single toss,

$$P(H) = \frac{1}{2}, \quad P(\overline{H}) = \frac{1}{2}.$$

Let A_k, B_k, C_k denote a head obtained by A, B, C respectively on the k th round. Since A starts,

$$\begin{aligned} P(\text{winning chance of } A) &= P\left(A_1 \cup (\overline{A_1} \cap \overline{B_1} \cap \overline{C_1} \cap A_2) \cup (\overline{A_1} \cap \overline{B_1} \cap \overline{C_1} \cap \overline{A_2} \cap \overline{B_2} \cap \overline{C_2} \cap A_3) \cup \dots\right) \\ &= \frac{1}{2} + \left(\frac{1}{2}\right)^3 \frac{1}{2} + \left(\frac{1}{2}\right)^6 \frac{1}{2} + \dots \\ &= \frac{1}{2} \left[1 + \frac{1}{8} + \left(\frac{1}{8}\right)^2 + \dots\right] = \frac{1}{2} \left(1 - \frac{1}{8}\right)^{-1} = \frac{1}{2} \times \frac{8}{7} = \boxed{\frac{4}{7}} \end{aligned}$$

Similarly,

$$P(\text{winning chance of } B) = \left(\frac{1}{2}\right) \frac{1}{2} \left(1 - \frac{1}{8}\right)^{-1} = \frac{1}{4} \times \frac{8}{7} = \boxed{\frac{2}{7}}$$

$$P(\text{winning chance of } C) = \left(\frac{1}{2}\right)^2 \frac{1}{2} \left(1 - \frac{1}{8}\right)^{-1} = \frac{1}{8} \times \frac{8}{7} = \boxed{\frac{1}{7}}$$

Check: $\frac{4}{7} + \frac{2}{7} + \frac{1}{7} = 1$. ✓

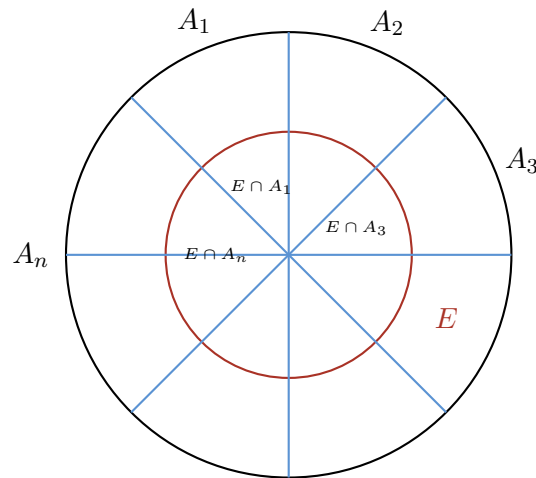
6 Bayes' Theorem

Definition 6.1 (Bayes' Theorem)

If $A_1, A_2, \dots, A_n$ is a set of n **mutually exclusive and exhaustive** events and E is an event associated with each A_i ($i = 1, 2, \dots, n$), then

$$P(A_i / E) = \frac{P(A_i) \cdot P(E / A_i)}{\sum_{i=1}^n P(A_i) \cdot P(E / A_i)}$$

It is also called the *Theorem of total probability*, or the *Theorem of probability of causes*.



Proof of Bayes' Theorem

Given that $A_1, A_2, \dots, A_n$ are a set of n mutually exclusive and exhaustive events, and E is an event associated with each A_i ($i = 1, 2, \dots, n$).

Since the A_i are mutually exclusive, the events $E \cap A_1, E \cap A_2, \dots, E \cap A_n$ are also mutually exclusive, and from the figure we see that

$$E = (E \cap A_1) \cup (E \cap A_2) \cup \dots \cup (E \cap A_n).$$

Therefore

$$P(E) = P(E \cap A_1) + P(E \cap A_2) + \dots + P(E \cap A_n),$$

that is,

$$P(E) = \sum_{i=1}^n P(E \cap A_i). \quad (2)$$

By the definition of conditional probability,

$$P(A_i / E) = \frac{P(A_i \cap E)}{P(E)} = \frac{P(A_i \cap E)}{\sum_{i=1}^n P(A_i \cap E)} \quad (\because \text{by (2)}).$$

Applying the multiplication theorem $P(A \cap B) = P(A) \times P(B / A)$ to both the numerator and the denominator gives

$$P(A_i / E) = \frac{P(A_i) \times P(E / A_i)}{\sum_{i=1}^n (P(A_i) \times P(E / A_i))}. \quad \blacksquare$$

Result 6.1 (Theorem of Total Probability)

By (2),

$$P(E) = \sum_{i=1}^n P(A_i \cap E) = \sum_{i=1}^n P(A_i) \times P(E / A_i)$$

is called the **total probability** for the event E .

Problem 8 (Packets routed through Mumbai)

Due to an internet configuration error, packets sent from Delhi to Chennai are routed through Mumbai with probability $3/4$. The probability that a packet routed through Mumbai is dropped is $1/3$, and the probability that a packet not routed through Mumbai is dropped is $1/4$. Find

- the probability that the packet is dropped;
- the conditional probability that a packet is routed through Mumbai, given that it is dropped;
- the conditional probability that a packet is routed through Mumbai, given that it is *not* dropped.

Solution 8

From the given statement we have the following events:

- A_1 — packets routed through Mumbai,

- A_2 — packets not routed through Mumbai,
- E — packets being dropped.

Hence

$$P(A_1) = \frac{3}{4}, \quad P(A_2) = 1 - P(A_1) = \frac{1}{4},$$

$$P(E / A_1) = \frac{1}{3}, \quad P(E / A_2) = \frac{1}{4}.$$

(i) **The packet is dropped.** By the theorem of total probability,

$$P(\text{packet is dropped}) = P(E) = (P(A_1) \times P(E / A_1)) + (P(A_2) \times P(E / A_2))$$

$$= \left(\frac{3}{4} \times \frac{1}{3}\right) + \left(\frac{1}{4} \times \frac{1}{4}\right) = \frac{1}{4} + \frac{1}{16} = \frac{5}{16}$$

(ii) **Routed through Mumbai, given that it is dropped.** By Bayes' theorem,

$$P(A_1 / E) = \frac{P(A_1) \times P(E / A_1)}{P(E)} = \frac{\frac{3}{4} \times \frac{1}{3}}{\frac{5}{16}} = \frac{1/4}{5/16} = \frac{4}{5}$$

(iii) **Routed through Mumbai, given that it is *not* dropped.**

$$P(A_1 / \bar{E}) = \frac{P(A_1) \times P(\bar{E} / A_1)}{(P(A_1) \times P(\bar{E} / A_1)) + (P(A_2) \times P(\bar{E} / A_2))}$$

$$= \frac{\frac{3}{4}(1 - P(E / A_1))}{\frac{3}{4}(1 - P(E / A_1)) + \frac{1}{4}(1 - P(E / A_2))} = \frac{\frac{3}{4}\left(1 - \frac{1}{3}\right)}{\frac{3}{4}\left(1 - \frac{1}{3}\right) + \frac{1}{4}\left(1 - \frac{1}{4}\right)}$$

$$= \frac{1/2}{\frac{1}{2} + \frac{3}{16}} = \frac{8}{11}$$

Problem 9 (Binary communication channel)

For a certain binary communication channel, the probability that a transmitted '0' is received as '0' is 0.95, and the probability that a transmitted '1' is received as '1' is 0.90. If the probability that a '0' is transmitted is 0.4, find the probability that

- a '1' is received;
- a '1' was transmitted, given that a '1' was received.

Solution 9

Define the events

- A_1 — transmitting '0',
- A_2 — transmitting '1',
- E — receiving '1'.

Then

$$P(A_1) = 0.4, \quad P(A_2) = 1 - 0.4 = 0.6,$$

$$P(E / A_1) = 1 - 0.95 = 0.05, \quad P(E / A_2) = 0.90.$$

(i) A '1' is received.

$$\begin{aligned} P(\text{a '1' is received}) &= P(E) = (P(A_1) \times P(E / A_1)) + (P(A_2) \times P(E / A_2)) \\ &= (0.4)(0.05) + (0.6)(0.90) \\ &= 0.02 + 0.54 = P(E) = 0.56 \end{aligned}$$

(ii) A '1' was transmitted, given that a '1' was received.

$$P(A_2 / E) = \frac{P(A_2) \times P(E / A_2)}{P(E)} = \frac{0.6(0.90)}{0.56} = \frac{0.54}{0.56} = P(A_2 / E) = 0.9643$$

Problem 10 (Bag of five items)

A bag contains 5 items and it is not known how many of them are white. Two items are drawn at random from the bag and they are noted to be white. What is the chance that all the items in the bag are white?

Solution 10

Since the number of white items is unknown, we treat every possibility as equally likely. Let

A_i = the event that the bag contains exactly i white items, $i = 0, 1, 2, 3, 4, 5$.

These are mutually exclusive and exhaustive, so

$$P(A_i) = \frac{1}{6} \quad (i = 0, 1, \dots, 5).$$

Let E be the event that the two items drawn are both white. Given that the bag has i white items,

$$P(E / A_i) = \frac{{}^iC_2}{{}^5C_2} = \frac{{}^iC_2}{10}.$$

Hence

$$P(E / A_0) = 0, \quad P(E / A_1) = 0, \quad P(E / A_2) = \frac{1}{10}, \quad P(E / A_3) = \frac{3}{10}, \quad P(E / A_4) = \frac{6}{10}, \quad P(E / A_5) = \frac{10}{10}$$

By the theorem of total probability,

$$P(E) = \sum_{i=0}^5 P(A_i) P(E / A_i) = \frac{1}{6} \left(0 + 0 + \frac{1}{10} + \frac{3}{10} + \frac{6}{10} + \frac{10}{10} \right) = \frac{1}{6} \times \frac{20}{10} = \frac{2}{6} = \frac{1}{3}.$$

Therefore, by Bayes' theorem,

$$\begin{aligned} P(\text{all the items in the bag are white}) &= P(A_5 / E) = \frac{P(A_5) \times P(E / A_5)}{P(E)} \\ &= \frac{\frac{1}{6} \times 1}{\frac{1}{3}} = \frac{1}{2} \end{aligned}$$

Problem 11 (Three machines producing washers)

Three machines A , B and C of equal capacities are producing washers. The probabilities that these machines produce defective washers are 0.15, 0.24 and 0.18 respectively. A washer taken out from a day's production was found to be defective. Find the probability that it was manufactured by machine C .

Solution 11

From the given statement we have the following events:

- A_1 — production of capacity of machine A ,
- A_2 — production of capacity of machine B ,
- A_3 — production of capacity of machine C ,
- E — production of defective washers.

Given that the three machines are of *equal capacities*,

$$P(A_1) = \frac{1}{3}, \quad P(A_2) = \frac{1}{3}, \quad P(A_3) = \frac{1}{3}.$$

Now

$$P(E / A_1) = 0.15, \quad P(E / A_2) = 0.24, \quad P(E / A_3) = 0.18.$$

Given that a washer taken out is found to be defective, by Bayes' theorem

$$\begin{aligned} P(\text{it was manufactured by machine } C) &= P(A_3 / E) = \frac{P(A_3) \times P(E / A_3)}{\sum_{i=1}^3 (P(A_i) \times P(E / A_i))} \\ &= \frac{P(A_3) \times P(E / A_3)}{(P(A_1) P(E / A_1)) + (P(A_2) P(E / A_2)) + (P(A_3) P(E / A_3))} \\ &= \frac{\frac{1}{3} \times 0.18}{\frac{1}{3}(0.15) + \frac{1}{3}(0.24) + \frac{1}{3}(0.18)} = \frac{0.18}{0.57} \\ &= P(A_3 / E) = \frac{6}{19} = 0.3157 \end{aligned}$$

Quick Reference

All Key Formulae at a Glance

Concept	Formula
Classical probability	$P(A) = \frac{n(A)}{n(S)}$
Complement	$P(\bar{A}) = 1 - P(A)$
Addition (two events)	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$
Addition (three events)	$P(A \cup B \cup C) = \sum P(A) - \sum P(A \cap B) + P(A \cap B \cap C)$
MEE	$P(A \cap B) = 0, \quad P(A \cup B) = P(A) + P(B)$
Conditional probability	$P(B / A) = \frac{P(A \cap B)}{P(A)}$
Multiplication (two)	$P(A \cap B) = P(A) P(B / A) = P(B) P(A / B)$
Multiplication (three)	$P(A \cap B \cap C) = P(A) P(B / A) P(C / A \cap B)$
Independence	$P(A \cap B) = P(A) P(B)$
De Morgan	$\overline{A \cup B} = \bar{A} \cap \bar{B}, \quad \overline{A \cap B} = \bar{A} \cup \bar{B}$
Total probability	$P(E) = \sum_{i=1}^n P(A_i) P(E / A_i)$
Bayes' theorem	$P(A_i / E) = \frac{P(A_i) P(E / A_i)}{\sum_{i=1}^n P(A_i) P(E / A_i)}$
Combinations	${}^nC_r = \frac{n!}{(n-r)! r!}$
Geometric series	$1 + x + x^2 + \cdots = (1 - x)^{-1}, x < 1$

Answers to the Problems

No.	Topic	Answer
1	Addition theorem (COVID-19)	(i) 0.3 (ii) 0.4
2	Three students	(i) 3/32 (ii) 13/32 (iii) 29/32
3	Three defectives, no replacement	1/120
4	Ninth test is last defective	8/195
5	At least one chip of each colour	48/91
6	Alternate throws (dice)	A : 30/61, B : 31/61
7	Three players tossing a coin	4/7, 2/7, 1/7
8	Packets through Mumbai	(i) 5/16 (ii) 4/5 (iii) 8/11
9	Binary channel	(i) 0.56 (ii) 0.9643
10	Bag of five items	1/2
11	Three machines	6/19 = 0.3157

End of Notes

Statistical Methods

Probability and Random Processes — Unit I

Random Variables, Statistical Averages, Moment Generating Functions,
Binomial, Poisson and Normal Distributions

Consolidated Theory, Results and Fully Worked Problems

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These consolidated notes bring together the five lecture modules of Unit I: **Random Variables, Statistical Averages, Moment Generating Functions, Binomial and Poisson Distributions** and the **Normal Distribution** — with complete theory, standard results and fully worked solutions.

Contents

1	Random Variables	3
1.1	What Do We Mean by a Random Variable?	3
1.2	Classification	3
1.3	Probability Function (PDF)	3
1.4	Properties of a Probability Density Function	4
1.5	Cumulative Distribution Function	4
1.6	Worked Problems	4
1.7	Additional Problems — Bayes' Theorem	11
2	Statistical Averages: Mean, Variance and Standard Deviation	14
2.1	Mean Value (Expectation)	14
2.2	Variance and Standard Deviation	14
2.3	Useful Results	14
2.4	Worked Problems	14
3	Moment Generating Function (MGF)	18
3.1	Why It Generates Moments	18
3.2	Worked Problems	18

4	Some Special Distributions	21
5	Binomial Distribution	22
5.1	Theory	22
5.2	Standard Results	23
5.3	Worked Problems	24
6	Poisson Distribution	29
6.1	Theory	29
6.2	Standard Results	29
6.3	Worked Problems	31
6.4	Binomial vs. Poisson — At a Glance	34
7	Normal Distribution	35
7.1	Theory	35
7.2	Worked Problems	36
7.3	Summary of Answers — Normal Distribution	43
	Consolidated Formula Sheet — Unit I	44

1 Random Variables

1.1 What Do We Mean by a Random Variable?

- Experiments often have *non-numerical* outcomes.
- Example:** a box has 3 red, 5 black and 2 white chips; 2 items are drawn. The pair may be (i) both white, (ii) both red, (iii) both black, (iv) one white and one red, (v) one white and one black, (vi) one red and one black.
- We *assign real numbers* to such outcomes.

Definition — Random Variable

Assigning a real number to every outcome of a random experiment gives a **random variable**, denoted $X, Y, Z, \dots$ or $X_1, X_2, X_3, \dots$

In the example above, if X = number of white chips drawn, then X takes the values 0, 1, 2.

1.2 Classification

Definition — Discrete Random Variable (DRV)

Assumes a discrete set of values.

Example: the random variable X discussed above.

Definition — Continuous Random Variable (CRV)

Assumes values in an interval.

Example: the life length of an electrical good.

1.3 Probability Function (PDF)

DRV — Probability Distribution (mass) Function

In tabular form:

$X = x$	x_1	x_2	$\dots$	x_n
$P(X = x)$	$P(X = x_1)$	$P(X = x_2)$	$\dots$	$P(X = x_n)$

CRV — Probability Density Function

A curve, or a functional form, for instance

$$f(x) = e^{-x}, \quad x \geq 0.$$

1.4 Properties of a Probability Density Function

If X is a DRV

- (i) $f(x) \geq 0$
- (ii) $\sum f(x) = 1$

If X is a CRV

- (i) $f(x) \geq 0$
- (ii) $\int_{-\infty}^{\infty} f(x) dx = 1$

1.5 Cumulative Distribution Function

$$F(x) = P(X \leq x)$$

CDF from PDF

$$F(x) = \begin{cases} \sum_{x_i \leq x} P(X = x_i), & X \text{ a DRV} \\ \int_{-\infty}^x f(x) dx, & X \text{ a CRV} \end{cases}$$

PDF from CDF

$$f(x) = \begin{cases} F(x_i) - F(x_{i-1}), & X \text{ a DRV} \\ \frac{d}{dx} F(x), & X \text{ a CRV} \end{cases}$$

1.6 Worked Problems

Problem 1.1 — Probability Distribution of White Chips

A box contains 3 red and 2 white coloured chips and 2 chips are drawn. If X denotes the number of white chips drawn, determine the PDF of X .

Solution — Setting up

X = number of white chips drawn, so X takes the values

$$X = 0 \text{ (no white),} \quad X = 1 \text{ (one white),} \quad X = 2 \text{ (two white).}$$

The total number of ways of drawing 2 chips from 5 is ${}^5C_2 = 10$.

Solution — The probabilities

$$P(X = 0) = P(\text{both are red}) = \frac{{}^3C_2}{{}^5C_2} = \frac{3}{10},$$

$$P(X = 1) = P(\text{one white and one red}) = \frac{{}^2C_1 \times {}^3C_1}{{}^5C_2} = \frac{6}{10},$$

$$P(X = 2) = P(\text{both are white}) = \frac{{}^2C_2}{{}^5C_2} = \frac{1}{10}.$$

Solution — PDF of X (X is a DRV)

$X = x$	0	1	2
$P(X = x)$	$\frac{3}{10}$	$\frac{6}{10}$	$\frac{1}{10}$

$$\text{Check: } \frac{3}{10} + \frac{6}{10} + \frac{1}{10} = 1. \checkmark$$

Problem 1.2 — Time Spent on an Electronic Gadget

The length of time (in minutes) that a person spends on an electronic gadget is a random phenomenon with probability function

$$f(x) = \begin{cases} A e^{-x/5}, & x \geq 0 \\ 0, & \text{otherwise.} \end{cases}$$

- (i) Find the value of A .
- (ii) What is the probability that the person spends (a) more than 10 minutes, (b) less than 5 minutes, (c) between 5 and 10 minutes?

Solution — (i) The value of A

Since $f(x)$ is a pdf, it satisfies $\int_{-\infty}^{\infty} f(x) dx = 1$:

$$\begin{aligned} \left(\int_{-\infty}^0 + \int_0^{\infty} \right) f(x) dx &= 1 \\ 0 + \int_0^{\infty} A e^{-x/5} dx &= 1 \implies A \left[\frac{e^{-x/5}}{-1/5} \right]_0^{\infty} = 1 \\ -5A(0 - 1) &= 1 \end{aligned}$$

$$A = \frac{1}{5}$$

$$\text{Hence } f(x) = \begin{cases} \frac{1}{5} e^{-x/5}, & x \geq 0 \\ 0, & \text{otherwise.} \end{cases}$$

Solution — (ii)(a) more than 10 minutes

$$\begin{aligned} P(X > 10) &= \int_{10}^{\infty} \frac{1}{5} e^{-x/5} dx \\ &= \frac{1}{5} \left[\frac{e^{-x/5}}{-1/5} \right]_{10}^{\infty} \\ &= - \left(0 - e^{-2} \right) = \boxed{\frac{1}{e^2}} \end{aligned}$$

Solution — (ii)(b) less than 5 minutes

$$\begin{aligned} P(X < 5) &= \int_0^5 \frac{1}{5} e^{-x/5} dx \\ &= - \left(e^{-1} - 1 \right) \\ &= \boxed{1 - \frac{1}{e}} \end{aligned}$$

Solution — (ii)(c) between 5 and 10 minutes

$$\begin{aligned}
 P(5 < X < 10) &= \int_5^{10} f(x) dx = \int_5^{10} \frac{1}{5} e^{-x/5} dx \\
 &= -\left(e^{-2} - e^{-1}\right) = e^{-1} - e^{-2} = \frac{1}{e} - \frac{1}{e^2} = \boxed{\frac{e-1}{e^2}}
 \end{aligned}$$

Problem 1.3 — Determining a and b for a Given PDF

A continuous random variable X has the pdf $f(x) = 3x^2$, $0 \leq x \leq 1$. Find a and b such that

- (i) $P(X \leq a) = P(X > a)$,
- (ii) $P(X > b) = 0.05$.

Solution — (i) Finding a

$$\begin{aligned}
 P(X \leq a) &= P(X > a) \\
 \int_0^a f(x) dx &= \int_a^1 f(x) dx \\
 \int_0^a 3x^2 dx &= \int_a^1 3x^2 dx \\
 3 \left[\frac{x^3}{3} \right]_0^a &= 3 \left[\frac{x^3}{3} \right]_a^1 \\
 a^3 &= 1 - a^3 \implies 2a^3 = 1
 \end{aligned}$$

$$a = \left(\frac{1}{2}\right)^{1/3} = (0.5)^{1/3}$$

Solution — (ii) Finding b

$$\begin{aligned}
 P(X > b) &= 0.05 \\
 \int_b^1 f(x) dx &= 0.05 \implies \int_b^1 3x^2 dx = 0.05 \\
 \left[x^3 \right]_b^1 &= 0.05 \implies 1 - b^3 = 0.05 \\
 b^3 &= 0.95
 \end{aligned}$$

$$b = (0.95)^{1/3}$$

Answers: $a = (0.5)^{1/3}$, $b = (0.95)^{1/3}$.

Problem 1.4 — Discrete PDF Involving a Parameter k

A random variable X has the PDF

X	0	1	2	3	4	5	6	7
$f(x)$	0	k	$2k$	$2k$	$3k$	k^2	$2k^2$	$7k^2 + k$

Find (i) the value of k ; (ii) $P(X < 6)$; (iii) $P(X \geq 6)$; (iv) $P(0 < X < 6)$; (v) $P(X \geq 5 | X > 3)$; (vi) the CDF of X .

Solution — (i) The value of k

X is a DRV, so $\sum f(x) = 1$:

$$k + 2k + 2k + 3k + k^2 + 2k^2 + 7k^2 + k = 1$$

$$10k^2 + 9k = 1 \implies 10k^2 + 9k - 1 = 0$$

$$10k^2 + 10k - k - 1 = 0 \implies 10k(k + 1) - (k + 1) = 0$$

$$(10k - 1)(k + 1) = 0 \implies k = -1 \text{ or } k = \frac{1}{10}$$

Since $f(x) \geq 0$, k cannot be negative.

$$k = \frac{1}{10}$$

Solution — The PDF of X with $k = 1/10$

$X = x$	0	1	2	3	4	5	6	7
$P(X = x)$	0	$\frac{1}{10}$	$\frac{2}{10}$	$\frac{2}{10}$	$\frac{3}{10}$	$\frac{1}{100}$	$\frac{2}{100}$	$\frac{17}{100}$

$$(\text{Here } 7k^2 + k = \frac{7}{100} + \frac{10}{100} = \frac{17}{100}.)$$

Solution — (ii), (iii) and (iv)

$$(ii) P(X < 6) = 1 - P(X \geq 6) = 1 - (P(X = 6) + P(X = 7)) = 1 - \frac{19}{100} = \boxed{\frac{81}{100}}$$

$$(iii) P(X \geq 6) = P(X = 6) + P(X = 7) = \frac{2}{100} + \frac{17}{100} = \boxed{\frac{19}{100}}$$

$$(iv) P(0 < X < 6) = P(X = 1) + P(X = 2) + P(X = 3) + P(X = 4) + P(X = 5) \\ = \frac{8}{10} + \frac{1}{100} = \boxed{\frac{81}{100}}$$

Solution — (v) Conditional probability

Using $P(A | B) = \frac{P(A \cap B)}{P(B)}$ with $(X \geq 5) \cap (X > 3) = (X \geq 5)$:

$$\begin{aligned} P(X \geq 5 | X > 3) &= \frac{P(X \geq 5)}{P(X > 3)} = \frac{P(X = 5) + P(X = 6) + P(X = 7)}{1 - P(X \leq 3)} \\ &= \frac{\frac{1}{100} + \frac{2}{100} + \frac{17}{100}}{1 - \frac{5}{10}} = \frac{20/100}{1/2} = \frac{1/5}{1/2} = \boxed{\frac{2}{5}} \end{aligned}$$

Solution — (vi) CDF of X , $F(x) = P(X \leq x)$

$X = x$	Computation	$F(x) = P(X \leq x)$
0	0	0
1	$0 + \frac{1}{10}$	$\frac{1}{10}$
2	$\frac{1}{10} + \frac{2}{10}$	$\frac{3}{10}$
3	$\frac{3}{10} + \frac{2}{10}$	$\frac{5}{10}$
4	$\frac{5}{10} + \frac{3}{10}$	$\frac{8}{10}$
5	$\frac{8}{10} + \frac{1}{100}$	$\frac{81}{100}$
6	$\frac{81}{100} + \frac{2}{100}$	$\frac{83}{100}$
7	$\frac{83}{100} + \frac{17}{100}$	1

Problem 1.5 — PDF from a Given CDF, and Back Again

The CDF of a continuous random variable X is

$$F(x) = \begin{cases} 0, & x < 0 \\ x^2, & 0 \leq x \leq \frac{1}{2} \\ 1 - \frac{3}{25}(3-x)^2, & \frac{1}{2} \leq x < 3 \\ 1, & x \geq 3. \end{cases}$$

Determine the PDF of X , and conversely recover the CDF from it.

Solution — PDF from the CDF

For a CRV, $f(x) = \frac{d}{dx}F(x)$. Differentiating piecewise (note $\frac{d}{dx}[-\frac{3}{25}(3-x)^2] = \frac{6}{25}(3-x)$):

$$f(x) = \begin{cases} 0, & x < 0 \\ 2x, & 0 \leq x \leq \frac{1}{2} \\ \frac{6}{25}(3-x), & \frac{1}{2} \leq x < 3 \\ 0, & x \geq 3. \end{cases}$$

Solution — CDF from the PDF — cases $x < 0$ and $0 \leq x \leq \frac{1}{2}$

For $x < 0$:

$$F(x) = \int_{-\infty}^x 0 \, dx = 0.$$

For $0 \leq x \leq \frac{1}{2}$:

$$F(x) = \int_{-\infty}^0 0 \, dx + \int_0^x 2x \, dx = 2 \left[\frac{x^2}{2} \right]_0^x = x^2.$$

Solution — CDF from the PDF — case $\frac{1}{2} \leq x < 3$

$$\begin{aligned} F(x) &= \int_{-\infty}^0 0 \, dx + \int_0^{1/2} 2x \, dx + \int_{1/2}^x \frac{6}{25} (3-x) \, dx \\ &= \left[x^2 \right]_0^{1/2} + \frac{6}{25} \left[\frac{(3-x)^2}{-2} \right]_{1/2}^x \\ &= \frac{1}{4} - \frac{3}{25} \left[(3-x)^2 - \left(\frac{5}{2} \right)^2 \right] \\ &= \frac{1}{4} + \frac{3}{4} - \frac{3}{25} (3-x)^2 = 1 - \frac{3}{25} (3-x)^2. \end{aligned}$$

Solution — CDF from the PDF — case $x \geq 3$

$$\begin{aligned} F(x) &= \int_{-\infty}^0 0 \, dx + \int_0^{1/2} 2x \, dx + \int_{1/2}^3 \frac{6}{25} (3-x) \, dx + \int_3^x 0 \, dx \\ &= \frac{1}{4} + \frac{6}{25} \left[\frac{(3-x)^2}{-2} \right]_{1/2}^3 = \frac{1}{4} - \frac{3}{25} \left[0^2 - \left(\frac{5}{2} \right)^2 \right] \\ &= \frac{1}{4} - \frac{3}{25} \left(-\frac{25}{4} \right) = \frac{1}{4} + \frac{3}{4} = 1. \end{aligned}$$

Hence the obtained CDF and PDF are consistent. ✓

Problem 1.6 — PDF and CDF from Proportionality Conditions

A random variable X takes the values 1, 2, 3, 4 such that

$$2P(X=1) = 3P(X=2) = P(X=3) = 5P(X=4).$$

Find the PDF and the CDF of X .

Solution — Finding the common value k

Let the common value be k , i.e. $2P(X=1) = 3P(X=2) = P(X=3) = 5P(X=4) = k$. Then

$$P(X=1) = \frac{k}{2}, \quad P(X=2) = \frac{k}{3}, \quad P(X=3) = k, \quad P(X=4) = \frac{k}{5}.$$

Since X is a DRV, $\sum f(x) = 1$:

$$\frac{k}{2} + \frac{k}{3} + k + \frac{k}{5} = 1 \implies \left(\frac{15 + 10 + 30 + 6}{30} \right) k = 1$$

$$\frac{61k}{30} = 1$$

$$k = \frac{30}{61}$$

Solution — PDF and CDF of X

$X = x$	1	2	3	4
$P(X = x)$	$\frac{15}{61}$	$\frac{10}{61}$	$\frac{30}{61}$	$\frac{6}{61}$
$F(x) = P(X \leq x)$	$\frac{15}{61}$	$\frac{25}{61}$	$\frac{55}{61}$	1

Problem 1.7 — Verifying that a Function is a PDF

Show that

$$f(x) = \begin{cases} x e^{-x^2/2}, & x \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

is a probability density function.

To prove

Since x takes values in an interval, X is a CRV, so we must show (i) $f(x) \geq 0$ and (ii) $\int_{-\infty}^{\infty} f(x) dx = 1$.

Solution

(i) Clearly $f(x) = x e^{-x^2/2} \geq 0$ for all $x \geq 0$.

(ii)

$$\int_{-\infty}^{\infty} f(x) dx = \left(\int_{-\infty}^0 + \int_0^{\infty} \right) f(x) dx = 0 + \int_0^{\infty} x e^{-x^2/2} dx.$$

Put $\frac{x^2}{2} = t \implies \frac{2x dx}{2} = dt \implies x dx = dt$ (no change in the limits):

$$\int_0^{\infty} e^{-t} dt = \left[\frac{e^{-t}}{-1} \right]_0^{\infty} = -(0 - 1) = 1.$$

By (i) and (ii), $f(x)$ is a pdf of a CRV X . ■

1.7 Additional Problems — Bayes' Theorem

Problem 1.8 — All Items in the Bag are White

A bag contains 5 items and is equally likely to contain 2, 3, 4 or 5 white items. Two items are drawn at random and both turn out to be white. What is the probability that *all* the items in the bag are white?

Solution — The events

A_1 : the bag contains 2 white items

A_2 : the bag contains 3 white items

A_3 : the bag contains 4 white items

A_4 : the bag contains 5 white items

E : drawing of 2 white items

with $P(A_1) = P(A_2) = P(A_3) = P(A_4) = \frac{1}{4}$.

Solution — Conditional probabilities

$$P(E | A_1) = \frac{{}^2C_2}{{}^5C_2} = \frac{1}{10},$$

$$P(E | A_2) = \frac{{}^3C_2}{{}^5C_2} = \frac{3}{10},$$

$$P(E | A_3) = \frac{{}^4C_2}{{}^5C_2} = \frac{6}{10},$$

$$P(E | A_4) = \frac{{}^5C_2}{{}^5C_2} = 1.$$

Solution — Applying Bayes' theorem

$$\begin{aligned} P(\text{all the items in the bag are white}) &= P(A_4 | E) \\ &= \frac{P(A_4) P(E | A_4)}{P(A_1)P(E | A_1) + P(A_2)P(E | A_2) + P(A_3)P(E | A_3) + P(A_4)P(E | A_4)} \\ &= \frac{\frac{1}{4}(1)}{\frac{1}{4}\left(\frac{1}{10}\right) + \frac{1}{4}\left(\frac{3}{10}\right) + \frac{1}{4}\left(\frac{6}{10}\right) + \frac{1}{4}(1)} \end{aligned}$$

$$P(A_4 | E) = \frac{1}{2} = 50\%$$

Problem 1.9 — Liver Disease and Regular Drinkers

In a certain town, 25% of the people drink alcohol regularly. The probability of death due to liver disease is 6 times greater for a regular drinker than for a non-drinker. If the probability of death due to liver disease in the town is 0.005, what is the probability that a person who died of liver disease was a regular drinker?

Solution — The events

A_1 : person drinks regularly, $P(A_1) = 0.25$

A_2 : person does not drink regularly, $P(A_2) = 0.75$

E : death due to liver disease, $P(E) = 0.005$

Solution — Solution

Let $P(E | A_1) = 6 P(E | A_2) = k$, so $P(E | A_2) = \frac{k}{6}$. From the total probability rule,

$$P(E) = P(A_1)P(E | A_1) + P(A_2)P(E | A_2)$$

$$0.005 = 0.25k + 0.75 \cdot \frac{k}{6} = 0.375k \implies k = \frac{1}{75}$$

By Bayes' theorem,

$$P(A_1 | E) = \frac{P(A_1)P(E | A_1)}{P(E)} = \frac{0.25 \times \frac{1}{75}}{0.005}$$

$$P(A_1 | E) = \frac{2}{3}$$

Problem 1.10 — All Items in the Lot are Good

A lot of 5 items is equally likely to contain 0, 1, 2 or 3 defective items. Two items are drawn at random and both are found to be good. Find the probability that all the items in the lot are good.

Solution — The events

A_1 : 0 defective, 5 good $P(A_1) = \frac{1}{4}$

A_2 : 1 defective, 4 good $P(A_2) = \frac{1}{4}$

A_3 : 2 defective, 3 good $P(A_3) = \frac{1}{4}$

A_4 : 3 defective, 2 good $P(A_4) = \frac{1}{4}$

E : drawing of 2 good items

Solution — Conditional probabilities

$$P(E | A_1) = \frac{{}^5C_2}{{}^5C_2} = 1,$$

$$P(E | A_2) = \frac{{}^4C_2}{{}^5C_2} = \frac{6}{10},$$

$$P(E | A_3) = \frac{{}^3C_2}{{}^5C_2} = \frac{3}{10},$$

$$P(E | A_4) = \frac{{}^2C_2}{{}^5C_2} = \frac{1}{10}.$$

Solution — Applying Bayes' theorem

Given that E has happened,

$$\begin{aligned} P(\text{all the items in the lot are good}) &= P(A_1 | E) = \frac{P(A_1) P(E | A_1)}{\sum_{i=1}^4 P(A_i) P(E | A_i)} \\ &= \frac{\frac{1}{4}(1)}{\frac{1}{4}\left(1 + \frac{6}{10} + \frac{3}{10} + \frac{1}{10}\right)} \end{aligned}$$

$$P(A_1 | E) = \frac{1}{2}$$

2 Statistical Averages: Mean, Variance and Standard Deviation

$$E(X) \quad \text{Var}(X) \quad \sigma = \sqrt{\text{Var}(X)}$$

2.1 Mean Value (Expectation)

Let X be a random variable (r.v.) with pdf $f(x)$. Then its **mean value** is defined by

Definition — Mean / Expectation

$$E(X) = \bar{x} = \mu = \begin{cases} \sum x f(x), & \text{if } X \text{ is a DRV} \\ \int_{-\infty}^{\infty} x f(x) dx, & \text{if } X \text{ is a CRV} \end{cases}$$

This mean value is also known as the **expectation** or **expected value** of the r.v. X .

2.2 Variance and Standard Deviation

Definition — Variance

$$\text{Var}(X) = V(X) = E[(X - \bar{x})^2] = \begin{cases} \sum (x - \bar{x})^2 f(x), & \text{if } X \text{ is a DRV} \\ \int_{-\infty}^{\infty} (x - \bar{x})^2 f(x) dx, & \text{if } X \text{ is a CRV} \end{cases}$$

Standard Deviation

$$\text{S.D.} = \sigma = \sqrt{\text{Var}(X)} \quad \leftarrow \quad \text{measures how the values spread out}$$

2.3 Useful Results

Results

1. $E(aX) = a E(X)$, where a is a constant.
2. $\text{Var}(aX \pm b) = a^2 \text{Var}(X)$, where a and b are constants.
3. $\text{Var}(X) = E(X^2) - (E(X))^2$

2.4 Worked Problems

Problem 2.1 — Expected Number of Tosses

A coin is tossed until a tail appears. What is the expectation of the number of tosses?

Solution — Setting up the PDF

Let X be a r.v. that denotes the number of tosses required to get a tail. Then we have the following pdf:

$X = x$	1	2	3	4	...
$P(X = x) = f(x)$	$\frac{1}{2}$	$\frac{1}{2^2}$ (HT)	$\frac{1}{2^3}$ (HHT)	$\frac{1}{2^4}$	...

Solution — Computing $E(X)$

$$\begin{aligned}
 E(X) &= \sum_x x f(x) = 1\left(\frac{1}{2}\right) + 2\left(\frac{1}{2^2}\right) + 3\left(\frac{1}{2^3}\right) + \cdots \\
 &= \frac{1}{2}\left(1 + 2\left(\frac{1}{2}\right) + 3\left(\frac{1}{2}\right)^2 + 4\left(\frac{1}{2}\right)^3 + \cdots\right) \\
 &= \frac{1}{2}(1 + 2y + 3y^2 + \cdots), \quad \text{where } y = \frac{1}{2} \\
 &= \frac{1}{2}(1 - y)^{-2} = \frac{1}{2}\left(1 - \frac{1}{2}\right)^{-2} = \frac{1}{2}\left(\frac{1}{2}\right)^{-2} = \frac{1}{2} \cdot 2^2 = 2
 \end{aligned}$$

$$E(X) = 2$$

On average, we require **2 tosses** to get a tail.

Problem 2.2 — Cars per Household

A market researcher at a major automobile company classified households by car ownership. The relative frequencies of the households for each category of ownership are as follows.

Number of cars per household	0	1	2	3	4	5
Relative frequencies	0.10	0.30	0.40	0.12	0.06	0.02

- Show that the given table of relative frequencies represents the pdf of the r.v. X (the number of cars per household).
- Find the mean, variance and standard deviation of X .

Solution — (a) The table is a pdf, and (b) the mean

It is clear from the table that all the frequency values are ≥ 0 and also

$$\sum (\text{all frequency values}) = 0.10 + 0.30 + 0.40 + 0.12 + 0.06 + 0.02 = 1$$

$\therefore$ The given table is the **pdf of the r.v. X** .

$$\begin{aligned}
 \therefore \text{Mean} &= E(X) = \sum_x x f(x) \\
 &= 0(0.10) + 1(0.30) + 2(0.40) + 3(0.12) + 4(0.06) + 5(0.02)
 \end{aligned}$$

$$E(X) = 1.8 \Rightarrow \text{the average number of cars per household is 1.8}$$

Solution — (b) Variance and standard deviation

$$\begin{aligned}
 \text{Var}(X) &= E(X^2) - (E(X))^2 = \sum_x x^2 f(x) - (1.8)^2 \\
 &= [0^2(0.10) + 1^2(0.30) + 2^2(0.40) + 3^2(0.12) \\
 &\quad + 4^2(0.06) + 5^2(0.02)] - (1.8)^2 \\
 &= 4.44 - (1.8)^2 = 4.44 - 3.24
 \end{aligned}$$

$$\text{Var}(X) = 1.2 \quad \text{S.D.} = \sigma = \sqrt{1.2} \Rightarrow \sigma = 1.09$$

Problem 2.3 — Average Profit per SIM Card

A mobile shop is promoting a sale in which the purchaser of a SIM card can roll a die and then deduct a rupee from the retail price for each dot shown on the rolled die. It is equally likely that the die will come up any integer from 1 to 6. The shop owner pays Rs. 5.00 for each SIM card and then prices it Rs. 9.00. During this special promotion, what will be the shop's average profit per SIM card sold?

Solution — Setting up the PDF

Let X denote the selling price of a SIM card. Since the die shows 1, 2, 3, 4, 5, 6 (each with probability $\frac{1}{6}$), the price Rs. 9 is reduced by that many rupees:

die shows	(1	2	3	4	5	6)
$X = x$	8	7	6	5	4	3
$f(x)$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$

Solution — Computing the average profit

$$E(X) = \sum_x x f(x) = 8\left(\frac{1}{6}\right) + 7\left(\frac{1}{6}\right) + 6\left(\frac{1}{6}\right) + 5\left(\frac{1}{6}\right) + 4\left(\frac{1}{6}\right) + 3\left(\frac{1}{6}\right)$$

$$E(X) = 5.5$$

The shop owner bought each SIM card at Rs. 5, so

$$\text{Profit on selling each SIM card} = 5.5 - 5 = 0.50$$

$$\text{Average profit} = \text{Rs. } 0.50 \text{ (50 paise) per SIM card}$$

Problem 2.4 — Mean and Variance of a Continuous r.v.

Find the mean and variance of the r.v. X whose pdf is given by

$$f(x) = \begin{cases} \frac{x+2}{8}, & -1 < x < 1 \\ 0, & \text{elsewhere} \end{cases}$$

Solution — Mean

Since X is a CRV,

$$\begin{aligned} \text{Mean} = E(X) &= \int_{-\infty}^{\infty} x f(x) dx = \int_{-1}^1 x \left(\frac{x+2}{8} \right) dx \\ &= \frac{1}{8} \int_{-1}^1 (x^2 + 2x) dx = \frac{1}{8} \left[\left(\frac{x^3}{3} \right)_{-1}^1 + 2 \left(\frac{x^2}{2} \right)_{-1}^1 \right] \\ &= \frac{1}{8} \left[\left(\frac{1}{3} - \left(-\frac{1}{3} \right) \right) + (1 - 1) \right] = \frac{1}{8} \left[\frac{2}{3} \right] = \frac{1}{12} \end{aligned}$$

$$\text{Mean} = E(X) = \frac{1}{12}$$

Solution — Variance

$\text{Var}(X) = E(X^2) - (E(X))^2$. Now,

$$\begin{aligned} E(X^2) &= \int_{-1}^1 x^2 f(x) dx = \int_{-1}^1 x^2 \left(\frac{x+2}{8} \right) dx \\ &= \frac{1}{8} \int_{-1}^1 (x^3 + 2x^2) dx = \frac{1}{8} \left[\frac{x^4}{4} + \frac{2x^3}{3} \right]_{-1}^1 \\ &= \frac{1}{8} \left[\left(\frac{1}{4} - \frac{1}{4} \right) + \frac{2}{3} (1 - (-1)) \right] = \frac{1}{8} \left[\frac{4}{3} \right] = \frac{1}{6} \end{aligned}$$

$$\therefore \text{Var}(X) = E(X^2) - (E(X))^2 = \frac{1}{6} - \left(\frac{1}{12} \right)^2 = \frac{1}{6} - \frac{1}{144}$$

$$\text{Var}(X) = \frac{23}{144} \approx 0.159$$

3 Moment Generating Function (MGF)

Definition — Moment Generating Function

Moments can easily be generated by the MGF, which is defined by

$$M_X(t) = E(e^{tX}) = \begin{cases} \sum e^{tx} f(x), & \text{if } X \text{ is a DRV} \\ \int_{-\infty}^{\infty} e^{tx} f(x) dx, & \text{if } X \text{ is a CRV} \end{cases}$$

3.1 Why It Generates Moments

Expansion of $M_X(t)$

$$\begin{aligned} M_X(t) &= E(e^{tX}) \\ &= E\left(1 + \frac{tX}{1!} + \frac{(tX)^2}{2!} + \frac{(tX)^3}{3!} + \cdots + \frac{(tX)^r}{r!} + \cdots\right) \\ &= 1 + \frac{t}{1!}E(X) + \frac{t^2}{2!}E(X^2) + \frac{t^3}{3!}E(X^3) + \cdots + \frac{t^r}{r!}E(X^r) + \cdots \\ &= 1 + \frac{t}{1!}\mu'_1 + \frac{t^2}{2!}\mu'_2 + \cdots + \frac{t^r}{r!}\mu'_r + \cdots \quad (\because \mu'_r = E(X^r)) \end{aligned}$$

$\Rightarrow \mu'_r = r^{\text{th}}$ moment about the origin.

$$\mu'_r = \text{Coeff. of } \frac{t^r}{r!} \text{ in the expansion of } M_X(t)$$

Results

$$\mu'_r = \left\{ \frac{d^r}{dt^r} M_X(t) \right\}_{t=0} \quad \mu'_1 = \text{mean of } X \quad \text{Var}(X) = \mu'_2 - (\mu'_1)^2$$

3.2 Worked Problems

Problem 3.1 — MGF of a Geometric-Type Discrete r.v.

Find the MGF of the r.v. X whose pdf is given by

$$f(x) = \frac{1}{2^x}, \quad x = 1, 2, 3, \dots$$

and hence find the mean and variance of X .

Solution — The MGF

Given X is a DRV with pdf $f(x) = \frac{1}{2^x}$, $x = 1, 2, 3, \dots$

$$\begin{aligned}
 \therefore \text{MGF} = M_X(t) &= E(e^{tX}) = \sum_x e^{tx} f(x) = \sum_{x=1,2,3,\dots} e^{tx} \frac{1}{2^x} \\
 &= \sum_{x=1,2,3,\dots} \left(\frac{e^t}{2}\right)^x = \frac{e^t}{2} + \left(\frac{e^t}{2}\right)^2 + \left(\frac{e^t}{2}\right)^3 + \dots \\
 &= \frac{e^t}{2} \left(1 + \frac{e^t}{2} + \left(\frac{e^t}{2}\right)^2 + \dots\right) \\
 &= \frac{e^t}{2} (1 + y + y^2 + \dots), \quad \text{where } y = \frac{e^t}{2} \\
 &= \frac{e^t}{2} (1 - y)^{-1} = \frac{e^t}{2} \left(1 - \frac{e^t}{2}\right)^{-1} = \frac{e^t}{2} \left(\frac{2}{2 - e^t}\right)
 \end{aligned}$$

$$M_X(t) = \frac{e^t}{2 - e^t}$$

Solution — Deduction — Mean

$$\begin{aligned}
 \text{Mean} = \mu'_1 &= \left\{ \frac{d}{dt} M_X(t) \right\}_{t=0} = \left\{ \frac{d}{dt} \left(\frac{e^t}{2 - e^t} \right) \right\}_{t=0} \\
 &= \left\{ \frac{(2 - e^t)e^t - e^t(-e^t)}{(2 - e^t)^2} \right\}_{t=0} = \left\{ \frac{2e^t - e^{2t} + e^{2t}}{(2 - e^t)^2} \right\}_{t=0} \\
 &= \left\{ \frac{2e^t}{(2 - e^t)^2} \right\}_{t=0} = \frac{2}{(2 - 1)^2} = 2
 \end{aligned}$$

$$\text{Mean} = 2$$

Solution — Deduction — Variance

Variance = $\mu'_2 - (\mu'_1)^2$, and

$$\begin{aligned}
 \mu'_2 &= \left\{ \frac{d^2}{dt^2} M_X(t) \right\}_{t=0} = \left\{ \frac{d}{dt} \left(\frac{d}{dt} M_X(t) \right) \right\}_{t=0} \\
 &= \left\{ \frac{d}{dt} \left(\frac{2e^t}{(2 - e^t)^2} \right) \right\}_{t=0} = \left\{ \frac{(2 - e^t)^2 2e^t - 2e^t \cdot 2(2 - e^t)(-e^t)}{(2 - e^t)^4} \right\}_{t=0} \\
 &= \frac{2 + 4}{1} = 6
 \end{aligned}$$

$$\therefore \text{Variance} = \mu'_2 - (\mu'_1)^2 = 6 - (2)^2$$

$$\text{Variance} = 2$$

Problem 3.2 — MGF of an Exponential r.v.

Find the MGF, mean and variance of the CRV X with pdf

$$f(x) = \begin{cases} \frac{1}{5} e^{-x/5}, & x \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

Solution — The MGF

$$\begin{aligned} M_X(t) &= E(e^{tX}) = \int_0^\infty e^{tx} f(x) dx = \int_0^\infty e^{tx} \frac{1}{5} e^{-x/5} dx \\ &= \frac{1}{5} \int_0^\infty e^{-x(\frac{1}{5}-t)} dx = \frac{1}{5} \left[\frac{e^{-(\frac{1}{5}-t)x}}{-\left(\frac{1}{5}-t\right)} \right]_0^\infty \\ &= \frac{1}{5} \left[\frac{1}{-\left(\frac{1}{5}-t\right)} (0-1) \right] = \frac{1}{5} \cdot \frac{1}{\left(\frac{1-5t}{5}\right)} = \frac{1}{1-5t} \end{aligned}$$

$$M_X(t) = \frac{1}{1-5t}$$

Solution — Deductions — Mean and Variance

$$\begin{aligned} \text{Mean} = \mu'_1 &= \left\{ \frac{d}{dt} M_X(t) \right\}_{t=0} = \left\{ \frac{d}{dt} \left(\frac{1}{1-5t} \right) \right\}_{t=0} \\ &= \left\{ (-1)(1-5t)^{-2} (-5) \right\}_{t=0} = \left\{ \frac{5}{(1-5t)^2} \right\}_{t=0} = 5 \end{aligned}$$

$$\text{Mean} = \mu'_1 = 5$$

$$\begin{aligned} \mu'_2 &= \left\{ \frac{d^2}{dt^2} M_X(t) \right\}_{t=0} = \left\{ \frac{d}{dt} \left(\frac{5}{(1-5t)^2} \right) \right\}_{t=0} = 50 \\ \Rightarrow \text{Variance} &= \mu'_2 - (\mu'_1)^2 = 50 - (5)^2 = 25 \end{aligned}$$

$$\text{Variance} = 25$$

4 Some Special Distributions

Relative to some probability function:

Discrete

- Binomial distribution
- Poisson distribution

Continuous

- Normal (or) Gaussian distribution
- Gamma (or) Erlang distribution

The Bernoulli Variate

- Takes only two values $X = 0$ (failure) and $X = 1$ (success)
- $P(X = 1) = p$ and $P(X = 0) = q$
- PDF $f(x) = P(X = x) = p^x q^{1-x}$, $x = 0, 1$
- Repeated n times, the PDF is $f(x) = P(X = x) = \binom{n}{x} p^x q^{n-x}$, $x = 0, 1, 2, \dots, n$

PART I

BINOMIAL DISTRIBUTION

5 Binomial Distribution

5.1 Theory

Bernoulli random variable — step by step

- Suppose X is a discrete random variable (DRV) taking only two values

$$X = 0 \text{ (failure), } X = 1 \text{ (success)}$$

$X = x$	0	1
$P(X = x)$	q	p

- such that

$$P(X = 1) = p, \quad P(X = 0) = q, \quad p + q = 1$$

- and its p.d.f. is

$$P(X = x) = p^x q^{1-x}, \quad x = 0, 1$$

From Bernoulli trials to the binomial p.d.f.

- The random variable which takes only the two values 0 and 1 is called a **Bernoulli random variable**.
- If we repeat a Bernoulli trial ' n ' times, then the probability of getting ' x ' successes in n trials is given by the p.d.f. ${}^nC_x p^x q^{n-x}$.

Definition — Binomial Distribution

A discrete random variable X is said to follow the **Binomial distribution** if its p.d.f. is of the form

$$f(x) = P(X = x) = \begin{cases} {}^nC_x p^x q^{n-x}, & x = 0, 1, 2, \dots, n \\ 0, & \text{otherwise} \end{cases}$$

Notation

$$X \sim B(n, p)$$

where n and p are the **parameters** of the Binomial distribution.

Note

Putting $x = 0, 1, 2, \dots, n$ in the p.d.f. gives

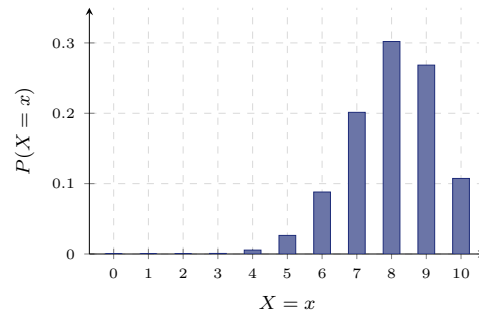
$$q^n, \quad {}^nC_1 p q^{n-1}, \quad {}^nC_2 p^2 q^{n-2}, \quad \dots, \quad p^n$$

which are the successive terms of the binomial expansion $(q+p)^n$ — hence the name **Binomial distribution**.

Shape of the binomial distribution

Example: $n = 10, p = 0.8$

- $P(X = x) = {}^{10}C_x (0.8)^x (0.2)^{10-x}$
- Total mass $= (q + p)^{10} = 1$
- Peak (mode) near $np = 8$
- Skewed to the left because $p > \frac{1}{2}$



Binomial distribution with $n = 10, p = 0.8$

5.2 Standard Results

Results for the Binomial Distribution

MGF: $M_X(t) = (q + pe^t)^n$

Mean: $\mu = np$

Variance: $\sigma^2 = npq$

Standard deviation: $\sigma = \sqrt{npq}$

Each of these is derived line by line below.

Result 1 — MGF of the binomial distribution is $(q + pe^t)^n$

Proof

Given $X \sim B(n, p)$, the p.d.f. is $f(x) = P(X = x) = {}^nC_x p^x q^{n-x}$, $x = 0, 1, 2, \dots, n$. Then

$$\begin{aligned}
 \text{MGF} &= M_X(t) = E(e^{tX}) = \sum_{x=0}^n e^{tx} f(x) = \sum_{x=0}^n e^{tx} {}^nC_x p^x q^{n-x} \\
 &= \sum_{x=0}^n {}^nC_x (pe^t)^x q^{n-x} \\
 &= q^n + {}^nC_1 (pe^t) q^{n-1} + {}^nC_2 (pe^t)^2 q^{n-2} + \dots + (pe^t)^n
 \end{aligned}$$

$$M_X(t) = (q + pe^t)^n$$

Reason used

$$(x + a)^n = x^n + {}^nC_1 x^{n-1} a + {}^nC_2 x^{n-2} a^2 + \dots + a^n$$

Deduction 1 — Mean of the binomial distribution

$$\begin{aligned}
 \text{Mean} &= \mu'_1 = \left\{ \frac{d}{dt} M_X(t) \right\}_{t=0} = \left\{ \frac{d}{dt} (q + pe^t)^n \right\}_{t=0} \\
 &= \left\{ n (q + pe^t)^{n-1} pe^t \right\}_{t=0} = n (q + p)^{n-1} p
 \end{aligned}$$

$$\text{Mean} = np$$

since $p + q = 1$

Deduction 2 — Variance of the binomial distribution

$$\text{Variance} = \mu'_2 - (\mu'_1)^2$$

$$\begin{aligned}\mu'_2 &= \left\{ \frac{d^2}{dt^2} M_X(t) \right\}_{t=0} = \left\{ \frac{d}{dt} \left(\frac{d}{dt} M_X(t) \right) \right\}_{t=0} \\ &= \left\{ \frac{d}{dt} \left(\underbrace{np e^t}_u \underbrace{(q + pe^t)^{n-1}}_v \right) \right\}_{t=0} \\ &= \left(np e^t (n-1) (q + pe^t)^{n-2} pe^t + (q + pe^t)^{n-1} np e^t \right)_{t=0}\end{aligned}$$

Putting $t = 0$ and using $q + p = 1$:

$$\mu'_2 = np(n-1)p + np = n^2p^2 - np^2 + np$$

$$\text{Variance} = \mu'_2 - (\mu'_1)^2 = n^2p^2 - np^2 + np - (np)^2 = -np^2 + np = np(1-p)$$

$$\text{Variance} = npq, \quad \text{S.D.} = \sqrt{npq}$$

since $p + q = 1 \Rightarrow 1 - p = q$

Summary Card — Binomial

Quantity	Binomial $X \sim B(n, p)$
p.d.f.	${}^nC_x p^x q^{n-x}, x = 0, 1, \dots, n$
MGF	$(q + pe^t)^n$
Mean	np
Variance	npq
S.D.	$\sqrt{npq}$
Condition	$p + q = 1$

5.3 Worked Problems

Problem 5.1 — Families with Four Children

Out of **800 families** with **4 children** each, how many families would be expected to have

- (i) 2 boys and 2 girls,
- (ii) at least 1 boy,
- (iii) at most 2 girls,
- (iv) children of both sexes?

Assume equal probabilities for boys and girls.

Solution — Setting up

Let X be the number of boys in a family.

$$P(\text{getting a boy}) = \frac{1}{2}; \quad P(\text{getting a girl}) = \frac{1}{2}$$

$$\Rightarrow p = \frac{1}{2}; \quad q = \frac{1}{2}$$

A family has 4 children $\Rightarrow n = 4$, so $X \sim B\left(4, \frac{1}{2}\right)$ and

$$P(X = x) = {}^4C_x \left(\frac{1}{2}\right)^x \left(\frac{1}{2}\right)^{4-x} = {}^4C_x \left(\frac{1}{2}\right)^4, \quad x = 0, 1, 2, 3, 4$$

$$P(X = x) = \frac{1}{16} {}^4C_x, \quad x = 0, 1, 2, 3, 4$$

x	0	1	2	3	4
4C_x	1	4	6	4	1
$P(X = x)$	$\frac{1}{16}$	$\frac{4}{16}$	$\frac{6}{16}$	$\frac{4}{16}$	$\frac{1}{16}$

Solution — (i) 2 boys and 2 girls

$$P(2 \text{ boys and 2 girls}) = P(X = 2) = \frac{1}{16} {}^4C_2 = \frac{6}{16} = \frac{3}{8}$$

$${}^4C_2 = \frac{4!}{2!2!} = 6$$

$\therefore$ Out of 800 families, the number having 2 boys and 2 girls

$$= \frac{3}{8} \times 800 = 300 \text{ families}$$

Solution — (ii) At least one boy

$$P(\text{at least one boy}) = P(X \geq 1) = 1 - P(X = 0) = 1 - \frac{1}{16} {}^4C_0 = \frac{15}{16}$$

$\therefore$ The number of families having at least one boy

$$= \frac{15}{16} \times 800 = 750 \text{ families}$$

Solution — (iii) At most 2 girls

Let Y denote the number of girls, so $Y = 4 - X$.

$$\begin{aligned} P(\text{at most 2 girls}) &= P(Y \leq 2) = P(Y = 0) + P(Y = 1) + P(Y = 2) \\ &= P(X = 4) + P(X = 3) + P(X = 2) \\ &= \frac{1}{16} {}^4C_4 + \frac{1}{16} {}^4C_3 + \frac{1}{16} {}^4C_2 = \frac{1 + 4 + 6}{16} = \frac{11}{16} \end{aligned}$$

$\therefore$ The number of families having at most 2 girls

$$= \frac{11}{16} \times 800 = 550 \text{ families}$$

Solution — (iv) Children of both sexes

$$\begin{aligned} P(\text{children of both sexes}) &= P(X = 1) + P(X = 2) + P(X = 3) \\ &= \frac{1}{16} {}^4C_1 + \frac{1}{16} {}^4C_2 + \frac{1}{16} {}^4C_3 = \frac{4 + 6 + 4}{16} = \frac{14}{16} = \frac{7}{8} \end{aligned}$$

$\therefore$ The number of families having children of both sexes

$$= \frac{7}{8} \times 800 = 700 \text{ families}$$

Problem 5.2 — Noisy Transmission Channel

A noisy transmission channel has a per digit error probability $p = 0.01$. Calculate the probability that **more than 1 error** occurs in **10 received digits**.

Solution — Setting up

Given $p = 0.01$, $n = 10$, so $q = 1 - p = 0.99$. Let X be the number of errors in the received digits; then $X \sim B(n, p)$ and

$$P(X = x) = {}^{10}C_x (0.01)^x (0.99)^{10-x}, \quad x = 0, 1, 2, \dots, 10.$$

Solution — Computation

$\therefore$ $P(\text{more than 1 error in 10 received digits})$

$$\begin{aligned} &= P(X > 1) = 1 - P(X \leq 1) = 1 - (P(X = 0) + P(X = 1)) \\ &= 1 - \left({}^{10}C_0 (0.01)^0 (0.99)^{10} + {}^{10}C_1 (0.01) (0.99)^9 \right) \\ &= 1 - \left((0.99)^{10} + 10 (0.01) (0.99)^9 \right) = 1 - (0.9044 + 0.0914) = 1 - 0.9957 \end{aligned}$$

$$P(X > 1) = 0.0043$$

Problem 5.3 — Finding the Parameter p

In a binomial distribution consisting of **5 independent trials**, the probabilities of **1** and **2** successes are 0.4096 and 0.2048 respectively. Find the parameter p of the distribution.

Solution — Setting up

Given $n = 5$ and $X \sim B(n, p)$, so $P(X = x) = {}^5C_x p^x q^{5-x}$, $x = 0, 1, 2, 3, 4, 5$. Also given

$$P(X = 1) = 0.4096 \quad \text{and} \quad P(X = 2) = 0.2048,$$

that is,

$${}^5C_1 p q^4 = 0.4096 \quad \text{and} \quad {}^5C_2 p^2 q^3 = 0.2048.$$

Solution — Solving for p

$$5p(1-p)^4 = 0.4096 \quad \text{and} \quad 10p^2(1-p)^3 = 0.2048$$

Dividing the first equation by the second:

$$\begin{aligned} \frac{5p(1-p)^4}{10p^2(1-p)^3} &= \frac{0.4096}{0.2048} \quad \Rightarrow \quad \frac{1-p}{2p} = 2 \\ \Rightarrow \quad 1-p &= 4p \quad \Rightarrow \quad 5p = 1 \end{aligned}$$

$$p = \frac{1}{5}$$

$${}^5C_1 = 5, \quad {}^5C_2 = \frac{5!}{2!3!} = 10$$

Problem 5.4 — Mean and Variance from the MGF

If the MGF of a random variable X is $(0.4e^t + 0.6)^8$, find its **mean** and **variance**.

Solution — Identifying the distribution

Given $M_X(t) = (0.4e^t + 0.6)^8$, which is of the form $(q + pe^t)^n$ — the MGF of a Binomial distribution. Hence

$$p = 0.4, \quad q = 0.6 \quad \text{and} \quad n = 8.$$

Using the standard results,

$$\text{Mean} = np = 8 \times 0.4 = 3.2$$

$$\text{Variance} = npq = 8 \times 0.4 \times 0.6 = 1.92$$

$$\text{Mean} = 3.2, \quad \text{Variance} = 1.92$$

Problem 5.5 — Basketball Free Throws

Over a long period of time, it has been observed that a professional basketball player can make a free throw on a given trial with probability **0.8**. Suppose he shoots **four free throws**.

- What is the probability that he will make **exactly two** free throws?
- What is the probability that he will make **at least one** free throw?

Solution — Setting up

Let X be the number of free throws made out of the four attempts. Given $p = 0.8$, $n = 4$, so $q = 1 - p = 0.2$. Each throw is an independent Bernoulli trial, hence $X \sim B(4, 0.8)$ and

$$P(X = x) = {}^4C_x(0.8)^x(0.2)^{4-x}, \quad x = 0, 1, 2, 3, 4.$$

Solution — (a) Exactly two free throws

$$P(X = 2) = {}^4C_2(0.8)^2(0.2)^2 = 6 \times (0.64) \times (0.04)$$

$$P(X = 2) = 0.1536$$

$${}^4C_2 = \frac{4!}{2!2!} = 6$$

Solution — (b) At least one free throw

$$P(X \geq 1) = 1 - P(X = 0) = 1 - {}^4C_0(0.8)^0(0.2)^4 = 1 - (0.2)^4 = 1 - 0.0016$$

$$P(X \geq 1) = 0.9984$$

$${}^4C_0 = 1 \quad \text{and} \quad (0.8)^0 = 1$$

PART II

POISSON DISTRIBUTION

6 Poisson Distribution

6.1 Theory

Poisson as the Limiting Case of the Binomial

$$P(X = x) = \lim_{\substack{n \rightarrow \infty \\ p \rightarrow 0 \\ (np = \lambda)}} {}^nC_x p^x q^{n-x} = \boxed{\frac{e^{-\lambda} \lambda^x}{x!}} \quad \text{where } \lambda = np$$

- Limiting case of the Binomial distribution
- $n \rightarrow \infty$, $p \rightarrow 0$, $np = \lambda$ a constant
- Apply when n is **large** and p is **small**

In the expansion one uses $n^x \left(1 - \frac{1}{n}\right) \left(1 - \frac{2}{n}\right) \cdots \rightarrow n^x$ as $n \rightarrow \infty$.

Definition — Poisson Distribution

A discrete random variable X is said to follow the **Poisson distribution** if its p.d.f. is of the form

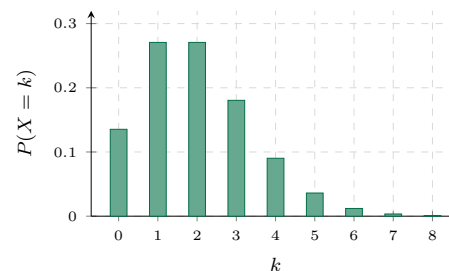
$$P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!}, \quad x = 0, 1, 2, \dots$$

Notation

$$X \sim P(\lambda)$$

where λ is the **parameter**; also

$$\boxed{\lambda = np}$$



Poisson distribution with $\lambda = 2$

6.2 Standard Results

Result — MGF of the Poisson distribution is $e^{\lambda(e^t - 1)}$

Proof

$$\begin{aligned}
 \text{MGF} &= M_X(t) = E(e^{tX}) = \sum_x e^{tx} f(x) = \sum_{x=0}^{\infty} e^{tx} \frac{e^{-\lambda} \lambda^x}{x!} \\
 &= e^{-\lambda} \sum_{x=0}^{\infty} \frac{(\lambda e^t)^x}{x!} \\
 &= e^{-\lambda} \left[1 + \frac{\lambda e^t}{1!} + \frac{(\lambda e^t)^2}{2!} + \frac{(\lambda e^t)^3}{3!} + \dots \right] = e^{-\lambda} e^{\lambda e^t}
 \end{aligned}$$

$$\text{MGF} = M_X(t) = e^{\lambda(e^t - 1)}$$

$$\text{using } e^y = 1 + \frac{y}{1!} + \frac{y^2}{2!} + \frac{y^3}{3!} + \dots$$

Deduction 1 — Mean of the Poisson distribution

$$\text{Mean} = \left\{ \frac{d}{dt} M_X(t) \right\}_{t=0} = \mu'_1 = \left\{ \frac{d}{dt} (e^{\lambda(e^t - 1)}) \right\}_{t=0} = \left\{ e^{\lambda(e^t - 1)} \cdot \lambda e^t \right\}_{t=0}$$

$$\text{Mean} = \lambda = \mu'_1$$

$$\text{at } t = 0: e^t = 1 \Rightarrow e^{\lambda(1-1)} = e^0 = 1$$

Deduction 2 — Variance of the Poisson distribution

$$\text{Variance} = \mu'_2 - (\mu'_1)^2$$

$$\begin{aligned}
 \mu'_2 &= \left\{ \frac{d^2}{dt^2} M_X(t) \right\}_{t=0} = \left\{ \frac{d}{dt} \left(\frac{d}{dt} M_X(t) \right) \right\}_{t=0} \\
 &= \left\{ \frac{d}{dt} \left(\underbrace{\lambda e^t}_u \underbrace{e^{\lambda(e^t - 1)}}_v \right) \right\}_{t=0} = \lambda(\lambda + 1) = \lambda^2 + \lambda
 \end{aligned}$$

$$\text{Variance} = \mu'_2 - (\mu'_1)^2 = \lambda^2 + \lambda - \lambda^2$$

$$\text{Variance} = \lambda$$

Characteristic property

$$\text{Mean} = \text{Variance} = \lambda$$

Summary Card — Poisson

Quantity	Poisson $X \sim P(\lambda)$
p.d.f.	$\frac{e^{-\lambda}\lambda^x}{x!}, x = 0, 1, 2, \dots$
MGF	$e^{\lambda(e^t-1)}$
Mean	λ
Variance	λ
S.D.	$\sqrt{\lambda}$
Link to Binomial	$\lambda = np$ (n large, p small)

6.3 Worked Problems

Problem 6.1 — Defective Pins

A manufacturer of pins knows that **2%** of his products are defective. If he sells pins in **boxes of 100** and guarantees that **not more than 4 pins** will be defective, what is the probability that a box will meet the guaranteed quality?

Solution — Setting up

Given $p = 0.02$ and $n = 100$; since n is large and p is small we may use the Poisson p.d.f. with

$$\lambda = np = (0.02) \times 100 = 2.$$

$$\lambda = 2$$

Let X be the number of defectives in a box, $X \sim P(\lambda)$; then

$$P(X = x) = \frac{e^{-\lambda}\lambda^x}{x!} = \frac{e^{-2}2^x}{x!}, \quad x = 0, 1, 2, 3, \dots$$

Solution — Computation

$\therefore P(\text{a box will fail to meet its guaranteed quality})$

$$\begin{aligned}
 &= P(X > 4) = 1 - P(X \leq 4) \\
 &= 1 - [P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3) + P(X = 4)] \\
 &= 1 - \left[\frac{e^{-2}2^0}{0!} + \frac{e^{-2}2^1}{1!} + \frac{e^{-2}2^2}{2!} + \frac{e^{-2}2^3}{3!} + \frac{e^{-2}2^4}{4!} \right] \\
 &= 1 - e^{-2} \left[1 + 2 + 2 + \frac{4}{3} + \frac{2}{3} \right] = 1 - e^{-2} [7]
 \end{aligned}$$

$$P(X > 4) = 0.0526$$

$\therefore$ The probability that a box *will* meet the guaranteed quality

$$= 1 - 0.0526 = 0.9474$$

Problem 6.2 — Car Hire Firm

A car hire firm has **two cars** which it hires out day by day. The number of demands for a car on each day is distributed as a **Poisson variate with mean 1.5**. Calculate the proportion of days on which

- (i) neither car is used,
- (ii) some demand is refused.

Solution — Setting up

Let X be the number of demands for a car on each day; it follows $P(\lambda)$ with mean $\lambda = 1.5$, so $X \sim P(1.5)$ and

$$P(X = x) = \frac{e^{-1.5}(1.5)^x}{x!}, \quad x = 0, 1, 2, \dots$$

But the firm has only two cars.

Solution — (i) Neither car is used

$$P(\text{neither car is used}) = P(X = 0) = \frac{e^{-1.5}(1.5)^0}{0!} = e^{-1.5}$$

$$P(X = 0) = 0.223$$

Solution — (ii) Some demand is refused

$$\begin{aligned} P(\text{some demand is refused}) &= P(X > 2) = 1 - P(X \leq 2) \\ &= 1 - [P(X = 0) + P(X = 1) + P(X = 2)] \\ &= 1 - \left[\frac{e^{-1.5}(1.5)^0}{0!} + \frac{e^{-1.5}(1.5)^1}{1!} + \frac{e^{-1.5}(1.5)^2}{2!} \right] \\ &= 1 - e^{-1.5} \left[1 + 1.5 + \frac{(1.5)^2}{2} \right] \end{aligned}$$

$$P(X > 2) = 0.1911$$

Problem 6.3 — Typographical Errors in a Book

In a book of **520 pages**, **390 typographical errors** occur. Assuming the Poisson law for the number of errors per page, find the probability that a **random sample of 5 pages** will contain **no error**.

Solution — Setting up

Let X be the number of errors per page; it follows $P(\lambda)$, where $\lambda = \text{mean} = \text{average error per page}$:

$$\lambda = \frac{390}{520} = \frac{3}{4} = 0.75 \quad \implies \quad X \sim P(0.75)$$

so that

$$P(X = x) = \frac{e^{-0.75}(0.75)^x}{x!}, \quad x = 0, 1, 2, 3, \dots$$

Recall $\lambda = np$ and $\lambda \rightarrow \text{mean}$.

Solution — Computation

$\therefore P(\text{a random sample of 5 pages will contain no error})$

$$\begin{aligned} &= P((X = 0) \cap (X = 0) \cap (X = 0) \cap (X = 0) \cap (X = 0)) \\ &= \underbrace{P(X = 0) \times P(X = 0) \times \dots \times P(X = 0)}_{5 \text{ times}} = (P(X = 0))^5 \\ &= \left(\frac{e^{-0.75}(0.75)^0}{0!} \right)^5 = (e^{-0.75})^5 \\ &= e^{-3.75} = 0.0235 \end{aligned}$$

Problem 6.4 — Traffic Accidents

The mean number of traffic accidents occurring on a specific stretch of highway is **two per week**. Assume the occurrence of accidents follows a Poisson distribution with parameter $\lambda = 2$.

- Calculate the probability of **no accidents** happening on this highway segment during a **1-week** period.
- Determine the probability of having **at most three accidents** on this highway segment during a **2-week** period.

Solution — Setting up

Let X be the number of accidents in a **1-week** period, $X \sim P(\lambda)$ with $\lambda = 2$:

$$P(X = x) = \frac{e^{-\lambda}\lambda^x}{x!} = \frac{e^{-2}2^x}{x!}, \quad x = 0, 1, 2, \dots$$

Key idea for part (b)

For a Poisson process the mean is **proportional to the length of the interval**. Over 2 weeks the parameter therefore becomes $\lambda' = 2 \times 2 = 4$.

Solution — (a) No accidents in a 1-week period

$$P(\text{no accidents in 1 week}) = P(X = 0) = \frac{e^{-2}2^0}{0!} = e^{-2}$$

$$P(X = 0) = 0.1353$$

since $2^0 = 1$ and $0! = 1$

Solution — (b) At most three accidents in 2 weeks

Let Y be the number of accidents in a 2-week period, so $Y \sim P(4)$.

$$\begin{aligned} P(Y \leq 3) &= P(Y = 0) + P(Y = 1) + P(Y = 2) + P(Y = 3) \\ &= e^{-4} \left[\frac{4^0}{0!} + \frac{4^1}{1!} + \frac{4^2}{2!} + \frac{4^3}{3!} \right] = e^{-4} \left[1 + 4 + 8 + \frac{32}{3} \right] = e^{-4} (23.6667) \end{aligned}$$

$$P(Y \leq 3) = 0.4335$$

6.4 Binomial vs. Poisson — At a Glance**Consolidated Comparison**

	Binomial $B(n, p)$	Poisson $P(\lambda)$
p.d.f.	${}^nC_x p^x q^{n-x}$	$\frac{e^{-\lambda} \lambda^x}{x!}$
Range of x	$0, 1, 2, \dots, n$ (finite)	$0, 1, 2, \dots$ (infinite)
Parameters	n, p	λ
MGF	$(q + pe^t)^n$	$e^{\lambda(e^t - 1)}$
Mean	np	λ
Variance	npq	λ
S.D.	$\sqrt{npq}$	$\sqrt{\lambda}$
Mean vs. Variance	Mean > Variance	Mean = Variance
Relation	$B(n, p) \rightarrow P(\lambda)$ as $n \rightarrow \infty, p \rightarrow 0, np = \lambda$	

PART III

NORMAL (GAUSSIAN) DISTRIBUTION

7 Normal Distribution

7.1 Theory

Binomial $\longrightarrow$ Normal distribution

In $B(n, p)$, when **neither p nor q is small**, i.e. np is large, Stirling's formula $n! = n^n e^{-n} \sqrt{2\pi n}$ gives

$$f(x) = \binom{n}{x} p^x q^{n-x} \approx \frac{1}{\sqrt{2\pi npq}} e^{-\frac{(x-np)^2}{2npq}}.$$

Since $\mu = np$ and $\sigma^2 = npq$,

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \quad P(x_1 \leq X \leq x_2) = \int_{x_1}^{x_2} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx$$

Definition — Normal (Gaussian) Distribution

A CRV X follows the **normal (Gaussian)** distribution if its PDF is

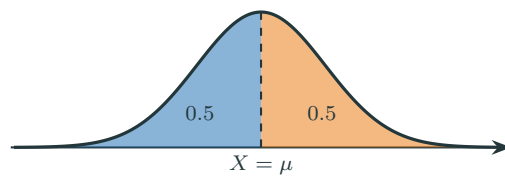
$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \quad -\infty < x < \infty$$

and we write $X \sim N(\mu, \sigma^2)$.

Results

(i) MGF = $e^{t\mu + \frac{\sigma^2 t^2}{2}}$ (ii) Mean = μ and Variance = σ^2

Properties of the normal curve



1. Symmetric about $X = \mu$ and bell shaped.
2. Total area under the curve is **unity**.
3. Area on either side of $X = \mu$ is **0.5**.

Standard normal variate

Let $X \sim N(\mu, \sigma^2)$ and put

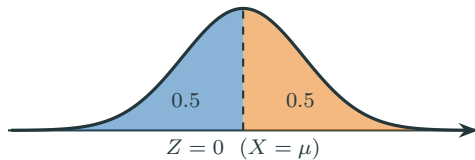
$$Z = \frac{X - \mu}{\sigma} \quad \implies \quad E(Z) = 0, \quad \text{Var}(Z) = 1$$

so that $Z \sim N(0, 1)$ is the **standard normal variate**, with PDF

$$\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}, \quad -\infty < z < \infty.$$

The curve is symmetric about $Z = 0$ and bell shaped.

Areas under the standard normal curve



$$P(-\infty < Z < 0) = 0.5$$

$$P(0 < Z < \infty) = 0.5$$

$$P(-\infty < Z < \infty) = 1$$

By symmetry,

$$P(-a < Z < 0) = P(0 < Z < a)$$

Normal tables give $P(0 < Z < a)$.

7.2 Worked Problems

Problem 7.1 — Heights of Soldiers

Assume that the mean height of soldiers is 68.22 inches with variance 10.88 inches. How many soldiers in a regiment of 2000 soldiers would you expect to be over 6 feet tall? Assume heights are normally distributed.

Solution — Step 1 — Data

Let $X \rightarrow$ height of soldiers, $X \sim N(\mu, \sigma^2)$

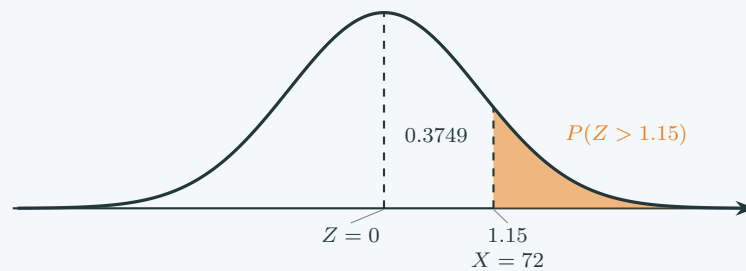
$$\mu = 68.22 \text{ inches}, \quad \sigma^2 = 10.88, \quad \sigma = \sqrt{10.88} = 3.2985$$

Since 6 feet = 72 inches, we need $P(\text{height more than 6 feet}) = P(X > 72)$.

Solution — Step 2 — Standardise

$$\begin{aligned} P(X > 72) &= P\left(\frac{X - \mu}{\sigma} > \frac{72 - \mu}{\sigma}\right) = P\left(Z > \frac{72 - 68.22}{\sqrt{10.88}}\right) = P\left(Z > \frac{3.78}{3.2985}\right) \\ &= P(Z > 1.146) \approx P(Z > 1.15) \end{aligned}$$

Solution — Step 3 — The normal curve



$$P(Z > 1.15) = 0.5 - P(0 < Z < 1.15) = 0.5 - 0.3749 = \mathbf{0.1251}$$

Solution — Step 4 — Answer

$$P(X > 6 \text{ feet}) = 0.1251$$

The number of soldiers, in a regiment of 2000, having height more than 6 feet

$$= 2000 \times 0.1251 = 250.2$$

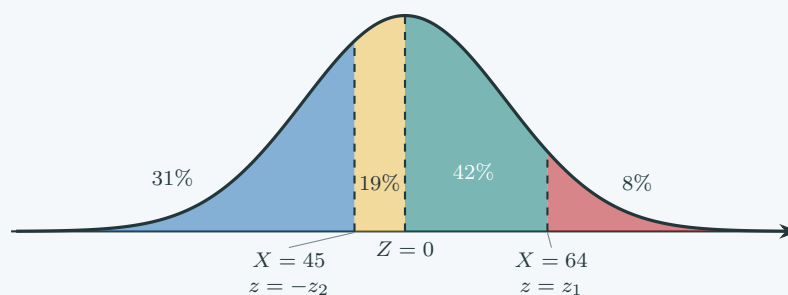
$$\approx 250 \text{ soldiers}$$

Problem 7.2 — Mean and S.D. from Percentages

In a normal distribution 31% of the items are under 45 and 8% of the items are over 64. Find the mean and standard deviation of the distribution.

Solution — Step 1 — The picture

Given $X \sim N(\mu, \sigma^2)$ with $P(X < 45) = 0.31$ and $P(X > 64) = 0.08$. Since $31\% < 50\%$, the value 45 lies to the **left** of μ ; since $8\% < 50\%$, the value 64 lies to the **right** of μ .

Solution — Step 2 — Read the z values

Let $Z = z_1$ and $Z = -z_2$ be the points corresponding to $X = 64$ and $X = 45$. From the figure:

$$P(0 < Z < z_1) = 42\% = 0.42 \implies \boxed{z_1 = 1.41}$$

$$P(-z_2 < Z < 0) = 19\% \implies P(0 < Z < z_2) = 0.19 \text{ (symmetry)} \implies \boxed{z_2 = 0.50}$$

(both from the normal table)

Solution — Step 3 — Two equations

Using $Z = \frac{X - \mu}{\sigma}$:

$$z_1 = \frac{64 - \mu}{\sigma} \quad 1.41 = \frac{64 - \mu}{\sigma}$$

$$\boxed{\mu + 1.41\sigma = 64} \quad (1)$$

$$-z_2 = \frac{45 - \mu}{\sigma} \quad -0.50 = \frac{45 - \mu}{\sigma}$$

$$\boxed{\mu - 0.50\sigma = 45} \quad (2)$$

Solution — Step 4 — Solve

(1) – (2):

$$1.91\sigma = 19 \implies \sigma = 9.9476$$

Substituting in (2):

$$\mu = 45 + 0.50(9.9476) = 45 + 4.974 = 49.97$$

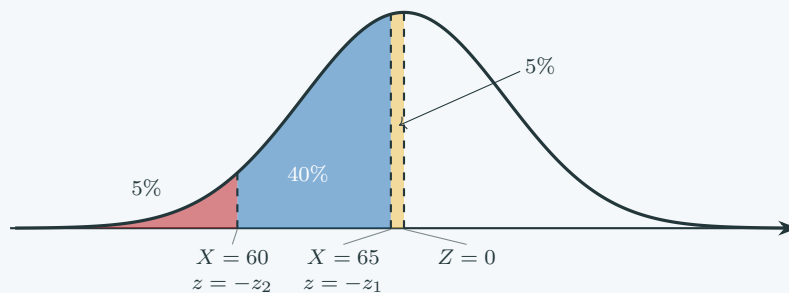
$$\text{Mean} = \mu \approx 50, \quad \text{S.D.} = \sigma \approx 10$$

Problem 7.3 — Heights of a Group of Men

Of a large group of men, 5% are under 60 inches in height and 40% are between 60 and 65 inches. Assuming a normal distribution, find the mean height and the standard deviation.

Solution — Step 1 — The picture

Given $P(X < 60) = 0.05$ and $P(60 < X < 65) = 0.40$, so $P(X < 65) = 0.45 < 0.5$. Hence **both** 60 and 65 lie to the left of μ .

**Solution — Step 2 — Read the z values**

Let $Z = -z_1$ and $Z = -z_2$ be the values corresponding to $X = 65$ and $X = 60$. From the figure:

$$\begin{aligned} P(-z_1 < Z < 0) &= 5\% \\ \Rightarrow P(0 < Z < z_1) &= 0.05 \\ \Rightarrow \boxed{z_1 = 0.13} \end{aligned}$$

$$\begin{aligned} P(-z_2 < Z < 0) &= 45\% \\ \Rightarrow P(0 < Z < z_2) &= 0.45 \\ \Rightarrow \boxed{z_2 = 1.65} \end{aligned}$$

Solution — Step 3 — Two equations

By the relation $Z = \frac{X - \mu}{\sigma}$:

$$-z_1 = \frac{65 - \mu}{\sigma} \quad -0.13 = \frac{65 - \mu}{\sigma}$$

$$\boxed{\mu - 0.13\sigma = 65} \quad (1)$$

$$-z_2 = \frac{60 - \mu}{\sigma} \quad -1.65 = \frac{60 - \mu}{\sigma}$$

$$\boxed{\mu - 1.65\sigma = 60} \quad (2)$$

Solution — Step 4 — Solve

(1) – (2):

$$1.52\sigma = 5 \implies \sigma = 3.289$$

Substituting in (1):

$$\mu = 65 + 0.13(3.289) = 65 + 0.4276 = 65.43$$

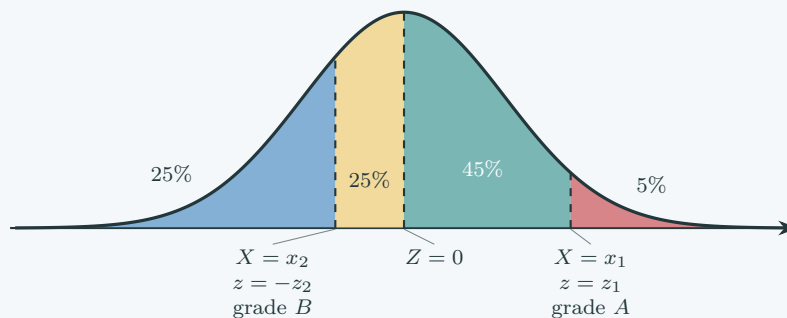
$$\text{Mean height} = 65.42'', \quad \text{S.D.} = 3.29''$$

Problem 7.4 — Examination Grades

A set of examination marks is approximately normally distributed with mean 75 and S.D. 5. If the top 5% of students get grade A and the bottom 25% get grade B, what mark is the lowest of A and what mark is the highest of B?

Solution — Step 1 — The picture

$X \rightarrow$ examination marks, $X \sim N(\mu, \sigma^2)$ with $\mu = 75$, $\sigma = 5$.

**Solution — Step 2 — Read the z values**

Let $X = x_1$ and $X = x_2$ be the lowest mark of grade A and the highest mark of grade B; let $Z = z_1$ and $Z = -z_2$ be the corresponding points.

$$P(0 < Z < z_1) = 45\% = 0.45 \implies \boxed{z_1 = 1.65}$$

$$P(-z_2 < Z < 0) = 25\% \implies P(0 < Z < z_2) = 0.25 \implies \boxed{z_2 = 0.68}$$

(both from the normal table)

Solution — Step 3 — Back-transform

Using $Z = \frac{X - \mu}{\sigma}$:

$$z_1 = \frac{x_1 - \mu}{\sigma} \quad 1.65 = \frac{x_1 - 75}{5}$$

$$x_1 = 75 + 8.25 = 83.25$$

$$-z_2 = \frac{x_2 - \mu}{\sigma} \quad -0.68 = \frac{x_2 - 75}{5}$$

$$x_2 = 75 - 3.40 = 71.60$$

Lowest mark of grade A ≈ 83 , Highest mark of grade B ≈ 72

Problem 7.5 — Examination Divisions

In an examination a student passes if he secures 30% or more marks. He is placed in the first, second and third divisions if he secures between 60% and 80%, between 45% and 60%, and between 30% and 45% respectively. He gets a distinction if he secures 80% or more. It is noticed that 10% of the students failed and 5% obtained a distinction. Calculate the percentage of students placed in the **second division**.

Solution — Step 1 — Data

Let $X \rightarrow$ marks obtained, $X \sim N(\mu, \sigma^2)$.

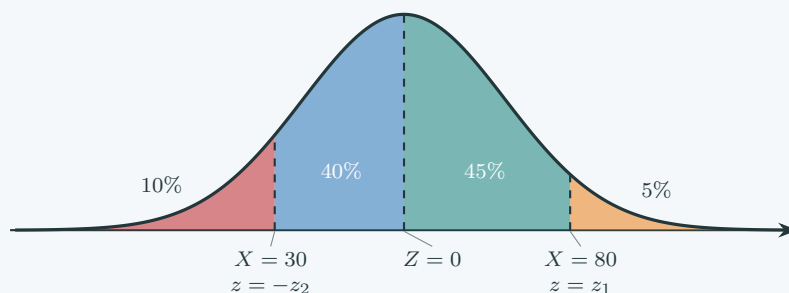
$$\text{Failed} \Rightarrow \text{marks below } 30 : \quad P(X < 30) = 10\% = 0.10$$

$$\text{Distinction} \Rightarrow \text{marks above } 80 : \quad P(X > 80) = 5\% = 0.05$$

Second division $\Rightarrow 45 < X < 60$ (this is what we must find).

Solution — Step 2 — The picture

Since $10\% < 50\%$, $X = 30$ lies to the left of μ ; since $5\% < 50\%$, $X = 80$ lies to the right of μ .

**Solution — Step 3 — Read the z values**

Let $Z = -z_2$ and $Z = z_1$ correspond to $X = 30$ and $X = 80$.

$$P(-z_2 < Z < 0) = 0.40$$

$$P(0 < Z < z_1) = 0.45$$

$$\Rightarrow P(0 < Z < z_2) = 0.40$$

$$\Rightarrow \boxed{z_2 = 1.28}$$

$$\Rightarrow \boxed{z_1 = 1.645}$$

(from the normal table)

Solution — Step 4 — Find μ and σ Using $Z = \frac{X - \mu}{\sigma}$:

$$-1.28 = \frac{30 - \mu}{\sigma}$$

$$\boxed{\mu - 1.28\sigma = 30} \quad (1)$$

$$1.645 = \frac{80 - \mu}{\sigma}$$

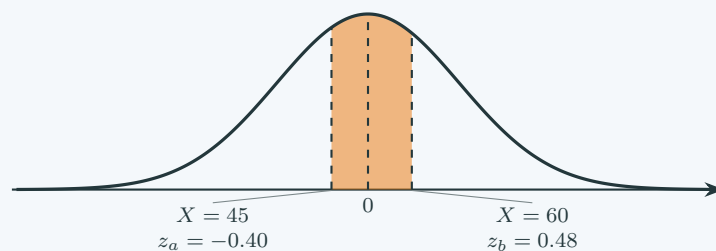
$$\boxed{\mu + 1.645\sigma = 80} \quad (2)$$

(2) - (1):

$$2.925\sigma = 50 \implies \sigma = 17.094 \quad \mu = 30 + 1.28(17.094) = 30 + 21.88 = 51.88$$

Solution — Step 5 — StandardisingSecond division $\Rightarrow P(45 < X < 60)$, with $\mu = 51.88$, $\sigma = 17.094$:

$$z_a = \frac{45 - 51.88}{17.094} = \frac{-6.88}{17.094} = -0.40, \quad z_b = \frac{60 - 51.88}{17.094} = \frac{8.12}{17.094} = 0.48$$

**Solution — Step 6 — Answer**

$$\begin{aligned} P(45 < X < 60) &= P(-0.40 < Z < 0.48) \\ &= P(-0.40 < Z < 0) + P(0 < Z < 0.48) \\ &= P(0 < Z < 0.40) + P(0 < Z < 0.48) \quad (\text{by symmetry}) \\ &= 0.1554 + 0.1844 = 0.3398 \end{aligned}$$

Students placed in the second division = $0.3398 \times 100 \approx 34\%$

Problem 7.6 — Life of Dry Battery Cells

A sample of 100 dry battery cells tested to find the length of life produced the following results:

$$\bar{x} = 12 \text{ hours}, \quad \sigma = 3 \text{ hours}$$

Assuming the data to be normally distributed, what percentage of battery cells are expected to have life

(i) more than 15 hours, (ii) less than 6 hours, (iii) between 10 and 14 hours?

(U.P. III Semester, Dec. 2003)

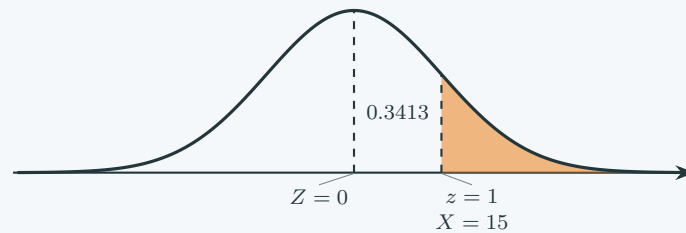
Solution — Step 1 — Data

Let $X \rightarrow$ length of life of a dry battery cell, $X \sim N(\mu, \sigma^2)$ with

$$\mu = \bar{x} = 12 \text{ hours}, \quad \sigma = 3 \text{ hours}, \quad Z = \frac{X - \mu}{\sigma} = \frac{X - 12}{3}.$$

Solution — (i) Life more than 15 hours

$$\text{When } X = 15 : \quad z = \frac{15 - 12}{3} = 1$$

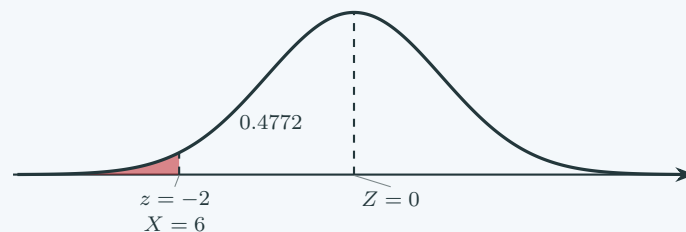


$$P(X > 15) = P(Z > 1) = 0.5 - P(0 < Z < 1) = 0.5 - 0.3413 = 0.1587$$

$\Rightarrow$ **15.87%** of the cells last more than 15 hours.

Solution — (ii) Life less than 6 hours

$$\text{When } X = 6 : \quad z = \frac{6 - 12}{3} = -2$$

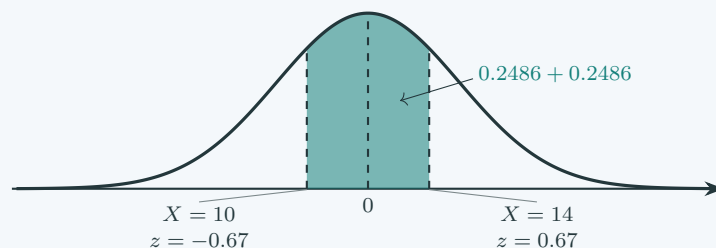


$$P(X < 6) = P(Z < -2) = 0.5 - P(-2 < Z < 0) = 0.5 - P(0 < Z < 2)$$

$$= 0.5 - 0.4772 = 0.0228 \quad \Rightarrow \quad \mathbf{2.28\%}$$

Solution — (iii) Life between 10 and 14 hours

$$X = 10 : z = \frac{10 - 12}{3} = -0.67 \quad \quad X = 14 : z = \frac{14 - 12}{3} = 0.67$$



$$\begin{aligned}
 P(10 < X < 14) &= P(-0.67 < Z < 0.67) \\
 &= P(-0.67 < Z < 0) + P(0 < Z < 0.67) \\
 &= 2P(0 < Z < 0.67) \quad (\text{by symmetry}) \\
 &= 2 \times 0.2486 = 0.4972
 \end{aligned}$$

49.72% of the battery cells have a life between 10 and 14 hours

7.3 Summary of Answers — Normal Distribution

Summary

No.	Problem	Answer
1	Soldiers over 6 feet (out of 2000)	≈ 250
2	Mean and S.D. (31% under 45, 8% over 64)	$\mu \approx 50, \sigma \approx 10$
3	Mean height and S.D. of men	$\mu = 65.42'', \sigma = 3.29''$
4	Lowest mark of A / highest mark of B	83 / 72
5	Students in the second division	$\approx 34\%$
6	Battery life: > 15h / < 6h / 10–14h	15.87% / 2.28% / 49.72%

Consolidated Formula Sheet — Unit I

All Distributions at a Glance

Quantity	Binomial	Poisson	Normal
	$B(n, p)$	$P(\lambda)$	$N(\mu, \sigma^2)$
p.d.f.	${}^nC_x p^x q^{n-x}$	$\frac{e^{-\lambda} \lambda^x}{x!}$	$\frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$
Type	Discrete	Discrete	Continuous
MGF	$(q + pe^t)^n$	$e^{\lambda(e^t-1)}$	$e^{\mu t + \frac{\sigma^2 t^2}{2}}$
Mean	np	λ	μ
Variance	npq	λ	σ^2
S.D.	$\sqrt{npq}$	$\sqrt{\lambda}$	σ

General Definitions

$$E(X) = \begin{cases} \sum x f(x) \\ \int_{-\infty}^{\infty} x f(x) dx \end{cases} \quad \text{Var}(X) = E(X^2) - (E(X))^2 \quad \sigma = \sqrt{\text{Var}(X)}$$

$$F(x) = P(X \leq x) \quad M_X(t) = E(e^{tX}) \quad \mu'_r = \left\{ \frac{d^r}{dt^r} M_X(t) \right\}_{t=0}$$

$$\text{Var}(X) = \mu'_2 - (\mu'_1)^2 \quad Z = \frac{X - \mu}{\sigma} \sim N(0, 1) \quad \lambda = np \quad (n \text{ large, } p \text{ small})$$

Thank You

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Gamma (or) Erlang Distribution

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Contents

1 The Gamma Integral	1
2 Definition	1
3 Moment Generating Function	2
4 Deduction of Mean and Variance	3
5 Problem 1 — Daily Milk Consumption	3
6 Problem 2 — Daily Electric Power Consumption	4
Summary	6

1 The Gamma Integral

Let

$$\Gamma(n) = \int_0^{\infty} e^{-x} x^{n-1} dx, \quad n > 0.$$

Substituting $x \rightarrow ax$ gives

$$\frac{\Gamma(n)}{a^n} = \int_0^{\infty} e^{-ax} x^{n-1} dx, \quad a > 0.$$

Hence

$$\int_0^{\infty} \frac{a^n e^{-ax} x^{n-1}}{\Gamma(n)} dx = 1.$$

Why this gives a density

The integrand

$$\frac{a^n e^{-ax} x^{n-1}}{\Gamma(n)}$$

is non-negative on $(0, \infty)$ and integrates to 1, so it is a valid probability density function.

Writing $a \rightarrow \lambda$ and $n \rightarrow k$ gives the Gamma density

$$f(x) = \frac{\lambda^k e^{-\lambda x} x^{k-1}}{\Gamma(k)}, \quad x \geq 0.$$

2 Definition

Definition

A continuous random variable X is said to follow the **Gamma (Erlang) distribution** if it takes the probability density function of the form

$$f(x) = \frac{\lambda^k e^{-\lambda x} x^{k-1}}{\Gamma(k)}, \quad 0 \leq x < \infty, \quad k > 0, \lambda > 0.$$

Here we write $X \sim \gamma(k, \lambda)$, where k and λ are the parameters of the distribution.

Results

For $X \sim \gamma(k, \lambda)$:

1. Moment generating function

$$M_X(t) = \left(\frac{\lambda}{\lambda - t} \right)^k, \quad t < \lambda$$

2. Mean = $\frac{k}{\lambda}$

3. Variance = $\frac{k}{\lambda^2}$

3 Moment Generating Function

$$\begin{aligned} M_X(t) &= E(e^{tx}) = \int_0^\infty e^{tx} f(x) dx \\ &= \int_0^\infty e^{tx} \frac{\lambda^k e^{-\lambda x} x^{k-1}}{\Gamma(k)} dx \\ &= \frac{\lambda^k}{\Gamma(k)} \int_0^\infty e^{-(\lambda-t)x} x^{k-1} dx. \end{aligned}$$

Using $\int_0^\infty e^{-ax} x^{n-1} dx = \frac{\Gamma(n)}{a^n}$ with $a = \lambda - t$ and $n = k$:

$$M_X(t) = \frac{\lambda^k}{\Gamma(k)} \cdot \frac{\Gamma(k)}{(\lambda - t)^k}.$$

MGF

$$M_X(t) = \left(\frac{\lambda}{\lambda - t} \right)^k, \quad t < \lambda.$$

4 Deduction of Mean and Variance

Mean

$$\begin{aligned} \mu'_1 &= \left\{ \frac{d}{dt} M_X(t) \right\}_{t=0} = \left\{ \frac{d}{dt} \lambda^k (\lambda - t)^{-k} \right\}_{t=0} \\ &= \left\{ \lambda^k (-k) (\lambda - t)^{-k-1} (-1) \right\}_{t=0} \\ &= k \lambda^k \lambda^{-k-1} = k \lambda^{-1}. \end{aligned}$$

Mean

$$\text{Mean} = \frac{k}{\lambda}$$

Variance

$$\text{Variance} = \mu'_2 - (\mu'_1)^2$$

$$\begin{aligned} \mu'_2 &= \left\{ \frac{d^2}{dt^2} M_X(t) \right\}_{t=0} = \left\{ \frac{d}{dt} \left(k \lambda^k (\lambda - t)^{-k-1} \right) \right\}_{t=0} \\ &= \left\{ k \lambda^k (-k-1) (\lambda - t)^{-k-2} (-1) \right\}_{t=0} \\ &= k(k+1) \lambda^k \lambda^{-k-2} = \frac{k^2}{\lambda^2} + \frac{k}{\lambda^2}. \end{aligned}$$

Therefore

$$\text{Variance} = \mu'_2 - (\mu'_1)^2 = \frac{k^2}{\lambda^2} + \frac{k}{\lambda^2} - \left(\frac{k}{\lambda} \right)^2.$$

Variance

$$\text{Variance} = \frac{k}{\lambda^2}$$

5 Problem 1 — Daily Milk Consumption

Problem 1

The daily consumption of milk in a city, in excess of 20,000 gallons, is approximately distributed as a Gamma variate with parameters $k = 2$ and $\lambda = \frac{1}{10,000}$. The city has a daily consumption stock of 30,000 gallons. What is the probability that the stock is insufficient on a particular day?

Solution — setting up

Let X be the daily consumption of milk in the city. Then $Y = X - 20,000$ is a random variable following the Gamma distribution with $k = 2$ and $\lambda = \frac{1}{10,000}$.

The p.d.f. of Y is

$$f(y) = \frac{\lambda^k e^{-\lambda y} y^{k-1}}{\Gamma(k)} = \frac{\left(\frac{1}{10000}\right)^2 e^{-\frac{y}{10000}} y^{2-1}}{\Gamma(2)}, \quad y > 0.$$

Since $\Gamma(2) = 1! = 1$,

p.d.f. of Y

$$f(y) = \frac{1}{(10000)^2} y e^{-\frac{y}{10000}}, \quad y > 0.$$

The city has a daily consumption stock of 30,000 gallons, so

$$P(\text{stock is insufficient on a particular day}) = P(X > 30,000).$$

$$\begin{aligned} P(X > 30000) &= P(X - 20000 > 30000 - 20000) = P(Y > 10000) \\ &= \int_{10000}^{\infty} f(y) dy = \int_{10000}^{\infty} \frac{1}{(10000)^2} y e^{-\frac{y}{10000}} dy. \end{aligned}$$

Put $\frac{y}{10000} = t \Rightarrow y = 10000 t, \quad dy = 10000 dt.$

When $y \rightarrow 10000$, $t \rightarrow 1$; when $y \rightarrow \infty$, $t \rightarrow \infty$. Hence

$$P(Y > 10000) = \frac{1}{(10000)^2} \int_1^{\infty} (10000) t e^{-t} (10000) dt = \int_1^{\infty} t e^{-t} dt.$$

Integrating by parts with $u = t$, $v = e^{-t}$ (Bernoulli's rule $uv_1 - u'v_2 + u''v_3 - \dots$):

$$\begin{aligned} \int_1^{\infty} t e^{-t} dt &= \left[t \frac{e^{-t}}{-1} - 1 \cdot \frac{e^{-t}}{(-1)^2} + 0 \right]_1^{\infty} \\ &= - (t e^{-t})_1^{\infty} - (e^{-t})_1^{\infty} \\ &= - (0 - e^{-1}) - (0 - e^{-1}) = 2e^{-1}. \end{aligned}$$

Answer

$$P(\text{stock is insufficient on a particular day}) = \frac{2}{e} \approx 0.7358.$$

On about 73.6% of days the stock of 30,000 gallons will be insufficient.

6 Problem 2 — Daily Electric Power Consumption

Problem 2

In a certain city, the daily consumption of electric power in millions of KWH can be treated as a random variable having a Gamma distribution with parameters $k = 3$ and $\lambda = \frac{1}{2}$. The power plant has a daily capacity of 12 million KWH. What is the probability that the power supply will be inadequate on a given day?

Solution — setting up

Let X be the daily consumption of electric power in millions of KWH. Given $X \sim \gamma(k, \lambda)$ with $k = 3$ and $\lambda = \frac{1}{2}$.

The p.d.f. of X is

$$\begin{aligned} f(x) &= \frac{1}{\Gamma(k)} \lambda^k e^{-\lambda x} x^{k-1}, \quad x > 0 \\ &= \frac{1}{\Gamma(3)} \left(\frac{1}{2}\right)^3 e^{-\frac{1}{2}x} x^{3-1} \quad [\Gamma(3) = 2! = 2] \\ &= \frac{1}{2} \cdot \frac{1}{8} x^2 e^{-x/2}. \end{aligned}$$

p.d.f. of X

$$f(x) = \frac{1}{16} x^2 e^{-x/2}, \quad x > 0.$$

Given the city has a daily capacity of 12 million KWH,

$$P(\text{power supply is inadequate on a day}) = P(X > 12).$$

$$P(X > 12) = \int_{12}^{\infty} f(x) dx = \frac{1}{16} \int_{12}^{\infty} \underbrace{x^2}_u \underbrace{e^{-x/2}}_v dx.$$

Apply Bernoulli's rule $uv_1 - u'v_2 + u''v_3 - \dots$ with $u = x^2$ and $v = e^{-x/2}$:

$$\begin{aligned}
P(X > 12) &= \frac{1}{16} \left[x^2 \frac{e^{-x/2}}{-\frac{1}{2}} - 2x \frac{e^{-x/2}}{\left(-\frac{1}{2}\right)^2} + 2 \frac{e^{-x/2}}{\left(-\frac{1}{2}\right)^3} \right]_{12}^{\infty} \\
&= \frac{1}{16} \left[-2 \left(x^2 e^{-x/2} \right)_{12}^{\infty} - 8 \left(x e^{-x/2} \right)_{12}^{\infty} - 16 \left(e^{-x/2} \right)_{12}^{\infty} \right] \\
&= \frac{1}{16} \left[-2 (0 - 144e^{-6}) - 8 (0 - 12e^{-6}) - 16 (0 - e^{-6}) \right] \\
&= \frac{1}{16} (288 + 96 + 16)e^{-6} = \frac{400}{16} e^{-6} = 25 e^{-6}.
\end{aligned}$$

Answer

$P(\text{power supply is inadequate on a given day}) = 25 e^{-6} \approx 0.0620$.

On about 6.2% of days the daily capacity of 12 million KWH will be inadequate.

Summary**Gamma / Erlang distribution — at a glance**

- PDF: $f(x) = \frac{\lambda^k e^{-\lambda x} x^{k-1}}{\Gamma(k)}, \quad x \geq 0, \quad k, \lambda > 0$
- MGF: $M_X(t) = \left(\frac{\lambda}{\lambda - t} \right)^k$
- Mean = $\frac{k}{\lambda}$, Variance = $\frac{k}{\lambda^2}$
- $k = 1$ gives the exponential distribution; integer k gives the Erlang distribution.